

HCSM-00 (Eleventh Revision): Foundations, Notation, and Mapping

The Hodge Complex Standard Model:
Substrate, Hodge Complex, Spectral Midpoint, Processor, Output Structures,
Emergent Physics, and Cosmology

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September 2026

HCSM White Paper Series, Paper 00 (Eleventh Revision)

Supersedes: HCSM-00 Tenth Revision

Abstract

The Hodge Complex Standard Model (HCSM) derives the gauge structure, matter content, particle spectrum, mixing matrices, spacetime dimension, graviton, Newton constant, and vacuum stability coefficient of the Standard Model from a single discrete substrate: the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. The substrate is selected by the unified substrate selection theorem. The emergent spacetime dimension is fixed by the topology of the Hodge-Dirac kernel. The Hodge complex of differential forms carries the operator $D = d + d^*$ with $D^2 = \Delta$ exactly on each form degree. The 14 modes at the self-paired eigenvalue $\lambda = 4$ are identified with one Standard Model generation.

This paper establishes the framework's foundational objects and states the notation. All rank blockers of the framework's derivation program are closed. The framework has a single dimensionful anchor (the Planck mass M_P) and derives every other quantity from the substrate at $d = 4$ and $L = 8$.

The present Eleventh Revision incorporates the full 2026 upgrade cycle. It integrates: the Tensor Product Theorem $V_{56} \cong V_{14} \otimes \ker D$ (HCSM-61) and its consequences for the continuum limit; the Maxwell-Lorentz sector (HCSM-61), Newtonian gravity (HCSM-62), first post-Newtonian corrections (HCSM-63), the relativistic energy-momentum limit (HCSM-64), the Dirac equation (HCSM-65), and direct quantization of the Hodge complex (HCSM-66); the loop and DNLS commutants (HCSM-15) and their extension to the full 56-dimensional Hodge sector (HCSM-16); the doubled-momentum pairing mechanism and exact $\mathfrak{su}(2)_L$ preservation; the dynamical bridge (HCSM-18); the Motion Theorem (HCSM-20); the native derivation of the $N/4$ labels from framework constants (HCSM-23 Rev. 8); the Orbit-Class Reconstruction and Distinguished-Pair Graph Decomposition (HCSM-28); the Charge-Entropy-Observer Triangle (HCSM-29); the Base-2 mass formula pattern (HCSM-30); the generation count and flavor mechanics (HCSM-31); the shell assignment rule (HCSM-32); the mass map with zero empirical anchors (HCSM-33 Rev. 2); the CKM matrix and its daughter structure (HCSM-34, HCSM-35); the PMNS matrix and its generation phase (HCSM-36, HCSM-37); the Higgs mechanism (HCSM-40); the absence of a light dark photon (HCSM-41); the spin-chirality $\mathbb{Z}_2$ grading (HCSM-42); the strong-CP solution from the orientation $\mathbb{Z}_2$ (HCSM-43); the Proca

sector (HCSM-44); the graviton as self-paired curvature mode (HCSM-45), its exact masslessness (HCSM-46), and its self-coupling (HCSM-47); the Faddeev-Popov normalization of the Newton constant (HCSM-48); the kL crossover (HCSM-51); the EFT coefficient in dark energy (HCSM-52); the dark energy density and DME entropy-well normalization (HCSM-53); vacuum stability and the stability coefficient (HCSM-54); the vacuum EFT (HCSM-56) and the Λ -promotion mechanism as pure projection (HCSM-56a).

The framework now has zero empirical anchors in the particle, gauge, hypercharge, mass, and dark energy sectors. Its single dimensionful anchor is M_P .

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Definition Index

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1 Introduction

1.1 What HCSM Is

The Hodge Complex Standard Model derives the Standard Model of particle physics from a single discrete substrate: the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. The framework's central claim is that the Standard Model is not a fundamental description of nature but the residue left when a small, discrete, information-bearing lattice cancels against its own mirror.

Physics is what the machine cannot cancel. The Hodge complex is the machine.

HCSM does not ask what fields exist or what symmetry groups are broken. It asks a single question: what substrate could produce exactly the Standard Model, and nothing else? The answer is a discrete lattice whose mirror cancellation leaves exactly the observed particle content. The mirror-cancellation theorem of HCSM-10 makes this statement precise: the Laplacian spectrum is exactly mirror-symmetric under $\lambda \leftrightarrow 8 - \lambda$, 50 of the 64 scalar modes pair into cancelling partners, and only the 14 self-paired modes at $\lambda = 4$ survive. The residual ratio is exactly $7/32$.

1.2 The Single Dimensionful Anchor

The framework has exactly one dimensionful input: the Planck mass M_P . Every other dimensionful quantity is a derived multiple of an appropriate power of M_P . The dimensionless integers $d = 4$ and $N = 64$ are derived from the substrate (Section 2); the substrate size $L = 8$ is derived from the unified selection theorem (Section 2.6); and the emergent Lorentzian signature $(3, 1)$ is derived from the Hodge star on $\ker D$ (Section 3, Theorem 3.9).

The electroweak scale is derived via

$$v_{\text{EW}} = M_P \exp(-kL_{\text{phys}}), \quad (1)$$

where $kL_{\text{phys}} = 38.442527$ is itself derived from the mass-map self-consistency condition (Section 8.1). The scale λ_L introduced in the gravity sector (Section 9.3) is derived as

$$\lambda_L = \frac{(d+1)M_P}{(d-1)L^3} = \frac{5M_P}{1536}, \quad (2)$$

reducing the formerly disclosed normalization $M_P^2 = NkL^2\lambda_L^2$ to a consistency relation.

At the particle level, the framework has zero empirical anchors. The identification of the 14 modes at $\lambda = 4$ with one Standard Model generation is a theorem (HCSM-23 Rev. 8). The charge table is a theorem whose binary residue is resolved by the same sorting theorem. The $N/4$ mass labels are derived from framework constants $\{4, 64, 8, 16\}$. The mass map coefficients are both derived.

As of the present revision, the gauge sector also has zero empirical anchors in the physical-identification step. The identification of the A_1 isotypic block with physical color $\mathfrak{su}(3)_c$ is a theorem (HCSM-06, Theorem 5.4). The identification of the B_1 block with physical weak isospin is a labeling convention (HCSM-06, Theorem 5.10), forced by the outer automorphism of D_4 .

1.3 Architectural Levels

The framework's architecture is a chain from the substrate to observable physics, organized in seven levels.

- Level 0: The Substrate.** The periodic torus $\Lambda_L = \mathbb{Z}_L \times \mathbb{Z}_L$ with the 5-point Laplacian. The substrate is selected by the unified substrate selection theorem from the single lattice primitive.
- Level 1: The Hodge Complex.** The exterior algebra $\Omega^0 + \Omega^1 + \Omega^2$ of differential forms with the Hodge-Dirac operator $D = d + d^*$, satisfying $D^2 = \Delta$ exactly on each form degree.
- Level 2: The Spectral Midpoint.** The self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian and its 14-mode eigenspace, carrying the D_4 isotypic decomposition $14 = 3A_1 + A_2 + 2B_1 + 2B_2 + 3E$.
- Level 3: The Processor and Its Dynamics.** The winding generator H_{wind} , the processor $T = I - P_C$, the class partition $\{A, B, C\}$, the loop operator M_{loop} , and the DNLS Hartree Hamiltonian $H_{\text{DNLS}} = 4I + UK$.
- Level 4: Output Structures.** The charge functional q_3 , the mass map $m_i = A_{\text{mass}} \exp(-B_{\text{mass}}(N/4)_i)$, the $N/4$ anchors, and the Midpoint Sorting Theorem.
- Level 5: Emergent Physics.** The gauge algebra $\mathfrak{g}_{\text{SM}} = \mathfrak{su}(3)_c + \mathfrak{su}(2)_L + \mathfrak{u}(1)^5$, hypercharge quantization, flavor mixing (CKM and PMNS), the Higgs sector, and gravity.
- Level 6: Cosmology.** The kL crossover, dark energy, vacuum stability, and the Λ -promotion mechanism.

1.4 Continuum Limit and Emergent Lorentz Invariance

The substrate is not a discretization of an underlying continuum. The framework's claim is the opposite: the continuum emerges from the substrate. The physical scale is set by the mass map and the kL crossover.

The 5-point Laplacian has the isotropic expansion

$$\lambda(n_1, n_2) = \frac{\pi^2}{16}(n_1^2 + n_2^2) + O(n^4). \quad (3)$$

As of the present revision, the continuum limit is closed through the Tensor Product Theorem $V_{56} \cong V_{14} \otimes \ker D$ (Section 9.1). The framework derives:

- The Maxwell-Lorentz sector, including the emergent Lorentzian metric of signature $(3, 1)$, the photon as the unbroken $\mathfrak{u}(1)_{\text{EM}}$ gauge boson, the Weinberg angle $\sin^2 \theta_W(M_Z) = 0.2312$, the W/Z masses and their ratio (Section 9.2).
- Newtonian gravity, the Poisson equation, the geodesic equation, and the equivalence principle (Section 9.3).
- First post-Newtonian corrections and the PPN parameters $\gamma = \beta = 1$, $\alpha_i = \zeta_i = \xi = 0$, $\eta_N = 0$ (Section 9.4).
- The relativistic energy-momentum relation $E^2 = p^2 c^2 + m^2 c^4$, the mass-shell condition $p^\mu p_\mu = -m^2 c^2$, and the Lorentz transformation properties of p^μ (Section 9.5).
- The Dirac equation from the Hodge-Dirac operator, including the Clifford algebra $\text{Cl}(3, 1)$, the chirality operator γ_5 , and the continuum limit $D/a \rightarrow i\gamma^\mu \partial_\mu$ (Section 9.6).
- Direct quantization of the Hodge complex, including the Fock space, propagator, path integral measure, and Ward identities (Section 9.7).

The remaining open items in the continuum sector are the explicit $O(\epsilon^4)$ corrections to the Yang-Mills action, the full non-Abelian Faddeev-Popov proof, the 2PN sector, and 1PN gravitational-wave propagation.

2 Level 0: The Substrate

2.1 The Periodic $L \times L$ Torus

Definition 2.1 (Periodic $L \times L$ Torus). The substrate is the periodic square lattice

$$\Lambda_L = \mathbb{Z}_L \times \mathbb{Z}_L, \quad N = L^2, \quad (4)$$

consisting of N sites arranged in an $L \times L$ grid with periodic boundary conditions. Site (x_1, x_2) is identified with $(x_1 + L, x_2)$ and with $(x_1, x_2 + L)$; equivalently, the coordinates x_1 and x_2 are each defined modulo L . Topologically, Λ_L is the torus $T^2 = S^1 \times S^1$.

The D_4 point group of the square acts on Λ_L by the eight rotations and reflections of the lattice. Its presentation is

$$D_4 = \langle r, s \mid r^4 = s^2 = (rs)^2 = e \rangle. \quad (5)$$

2.2 The 5-Point Laplacian Is Forced

Theorem 2.2 (Laplacian Uniqueness). *Let $\Delta : \mathbb{C}^N \rightarrow \mathbb{C}^N$ be a linear operator on the substrate $\Lambda_L = \mathbb{Z}_L \times \mathbb{Z}_L$ satisfying:*

1. *D_4 -invariance: $[R, \Delta] = 0$ for every $R \in D_4$;*
2. *nearest-neighbour locality: $(\Delta f)(x)$ depends only on $f(x)$ and f at the four nearest neighbours of x ;*
3. *positivity and kernel: Δ is positive-semidefinite and $\ker \Delta$ is exactly the set of constant functions.*

Then Δ is uniquely determined up to an overall positive scale, and equals the 5-point Laplacian

$$(\Delta f)(x_1, x_2) = 4f(x_1, x_2) - \sum_{|e|=1} f(x + e). \quad (6)$$

Proof. Condition (2) writes $\Delta f(x) = a_0 f(x) + \sum_{\mu} a_{\mu} f(x + \mu)$. The four nearest-neighbour directions $\pm e_1, \pm e_2$ lie in a single D_4 orbit. Hence condition (1) forces $a_{\mu} = a$ for all four directions. Condition (3) forces $\Delta 1 = 0$, so $a_0 + 4a = 0$, giving $a_0 = -4a$. Normalizing $a = 1$ gives the 5-point Laplacian. $\square$

Remark 2.3. The 5-point Laplacian annihilates the constants by construction. Its eigenvalue on the plane wave

$$\phi_{n_1, n_2}(x) = \exp(2\pi i(n_1 x_1 + n_2 x_2)/L) \quad (7)$$

is

$$\lambda(n_1, n_2) = 4 - 2\cos(2\pi n_1/L) - 2\cos(2\pi n_2/L), \quad (8)$$

with $n_1, n_2 \in \{0, 1, \dots, L-1\}$. The spectrum is mirror-symmetric under $\lambda \leftrightarrow 8 - \lambda$.

2.3 The Framework Relation $N = d \cdot 2^d$

The framework uses the relation

$$N = d \cdot 2^d, \quad (9)$$

with N the number of substrate sites and d the emergent spacetime dimension. At $d = 4$ this gives $N = 4 \cdot 2^4 = 64$. Since $\Lambda_L = \mathbb{Z}_L \times \mathbb{Z}_L$ requires $N = L^2$, this gives $L = \sqrt{64} = 8$.

At $d = 4$, $N = 64$ is a perfect square. The integer solutions of $d \cdot 2^d = L^2$ form the infinite family $d \in \{4k^2 : k \geq 1\}$, with $d = 16, 36, 64, \dots$ also giving perfect squares. The physical selection of $d = 4$ comes from the unified selection theorem of Section 2.6.

2.4 Two Derivations of $L = 8$

The substrate size $L = 8$ is selected by two independent derivations.

First derivation: equal entropy spacing. On the $L \times L$ torus, the standing-wave modes fall into discrete Shannon entropy classes.

Theorem 2.4 (Equal Entropy Spacing Selects $L = 8$). *Among all integers $4 \leq L \leq 10^4$, the only L for which the $L \times L$ torus has exactly three equally spaced Shannon entropy classes is $L = 8$. At $L = 8$ the three classes are 6.0 bits, 5.5 bits, and 5.0 bits, with uniform spacing 0.5 bits.*

Second derivation: maximal winding cancellation. The winding generator generalizes to any even L as $H_{\text{wind}}(L) = i \sin(\pi M/L)$ on the self-paired mode set of Λ_L .

Theorem 2.5 (Maximal Cancellation Selects $L = 8$). *Let G_L be the parity block graph of $H_{\text{wind}}(L)$ on the five nodes $\{A_{\text{odd}}, A_{\text{even}}, B_{\text{odd}}, B_{\text{even}}, AB\}$. The number of block couplings that vanish identically is*

$$\#\text{zeros}(G_L) = \begin{cases} 3 & L = 4, \\ 4 & L \equiv 2 \pmod{4}, \\ 7 & L \equiv 0 \pmod{4}, L \geq 12, \\ 10 & L = 8. \end{cases} \quad (10)$$

Among the two maximally-cancelling sizes, only $L = 8$ carries the 14-mode multiplet. At $L = 8$ the surviving five block couplings form a unicyclic graph on $d + 1 = 5$ nodes with rank-1 edges; the winding kernel has dimension $d(d + 1)/2 = 10$; and the kernel-diagonal values are $\{2/d, (d - 1)/d\}$.

The two derivations are independent: the first uses only the torus Laplacian's entropy classes; the second uses only the winding generator's parity block structure.

2.5 The 14-Mode Count

Theorem 2.6 (Midpoint Multiplicity). *On the $L \times L$ torus with L even, the multiplicity of the self-paired eigenvalue $\lambda = 4$ is*

$$\text{mult}_{\lambda=4}(L) = 2(L - 1). \quad (11)$$

At $L = 8$ this gives 14.

Proof. The self-paired eigenvalue is the fixed point of the mirror symmetry $\lambda \leftrightarrow 8 - \lambda$. Its eigenspace consists of the modes (n_1, n_2) satisfying $\cos(2\pi n_1/L) + \cos(2\pi n_2/L) = 0$. Using the sum-to-product identity, this becomes $2 \cos(\pi(n_1 + n_2)/L) \cos(\pi(n_1 - n_2)/L) = 0$. The solutions are $n_1 + n_2 = L/2 \pmod{L}$ or $n_1 - n_2 = L/2 \pmod{L}$. Each congruence has L solutions; the two congruences intersect in exactly two points. By inclusion-exclusion, the total is $2L - 2 = 2(L - 1)$. $\square$

The 14 modes at $\lambda = 4$ on $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ are the subject of HCSM-04 and are identified with one Standard Model generation in HCSM-23 Rev. 8.

2.6 Unified Substrate Selection

The substrate size $L = 8$ and the emergent dimension $d = 4$ are selected by four independent derivations in the framework: equal entropy spacing on the torus Laplacian; maximal winding cancellation of the winding generator; Lorentzian signature on $\ker D$; and the Betti numbers of T^2 . This subsection shows that they are corollaries of a single selection theorem.

2.6.1 The Two Framework Identities

Identity 1 (site-count relation). The number of substrate sites N and the emergent dimension d satisfy

$$N = d \cdot 2^d. \quad (12)$$

Identity 2 (mode-content relation). The self-paired mode count of the torus Laplacian at the midpoint eigenvalue equals the total particle content of one Standard Model generation:

$$\text{mult}_{\lambda=4}(L) = |F| + |B_{\text{bos}}|. \quad (13)$$

The right-hand side is derived content: the fermion-sector dimension $|F| = d(d+1)/2$ (HCSM-13, HCSM-19) and the boson-sector dimension $|B_{\text{bos}}| = d$ (HCSM-13, HCSM-19). At $d = 4$ they give $|F| = 10$, $|B_{\text{bos}}| = 4$, and total content 14.

The left-hand side is HCSM-04's multiplicity theorem: $\text{mult}_{\lambda=4}(L) = 2(L-1)$.

2.6.2 The Selection Theorem

Theorem 2.7 (Unified Substrate Selection). *The system*

$$L^2 = d \cdot 2^d, \quad 2(L-1) = \frac{d(d+3)}{2}, \quad (14)$$

has the unique even-integer solution $L = 8$, $d = 4$, $N = 64$.

Proof. By Identity 1, $L = \sqrt{d \cdot 2^d}$. For L to be an even integer, $d \cdot 2^d$ must be a perfect square. Since 2^d is a perfect square iff d is even, write $d = 2m$. Then $d \cdot 2^d = 2m \cdot 2^{2m}$, which is a perfect square iff $2m$ is a perfect square, i.e., $m = 2k^2$ for integer $k \geq 1$. Hence

$$d = 4k^2, \quad L = 2^{2k^2+1} \cdot k. \quad (15)$$

The mode-content identity gives

$$2^{2k^2+1}k - 1 = k^2(4k^2 + 3). \quad (16)$$

For $k = 1$: LHS = $2^3 \cdot 1 - 1 = 7$, RHS = $1 \cdot 7 = 7$. Satisfied. For $k = 2$: LHS = $2^9 \cdot 2 - 1 = 1023$, RHS = $4 \cdot 19 = 76$. LHS exceeds RHS. For $k \geq 3$: LHS grows super-exponentially in k^2 ; RHS grows polynomially. LHS strictly dominates for all $k \geq 2$. Hence $k = 1$ is the unique solution, giving $d = 4$, $L = 8$, $N = 64$. $\square$

Corollary 2.8 (Uniqueness). *The pair $(L, d) = (8, 4)$ is the unique even-integer pair satisfying the two framework identities simultaneously.*

2.6.3 The Four Prior Derivations as Corollaries

Corollary 2.9 (Entropy Spacing). *At $L = 8$, the substrate $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ has exactly three equally spaced Shannon entropy classes: 6.0, 5.5, and 5.0 bits.*

Corollary 2.10 (Maximal Winding Cancellation). *At $L = 8$, the winding generator $H_{\text{wind}} = i \sin(\pi M/8)$ has cancellation depth 10 on the five-node parity block graph, with kernel dimension $d(d+1)/2 = 10$.*

Corollary 2.11 (Lorentzian Signature). *At $L = 8$, the Hodge star on $\ker D = H^0 + H^1 + H^2$ has eigenvalues $\{+1, -1, +i, -i\}$, giving a metric of signature $(3, 1)$ on $\ker D$.*

Corollary 2.12 (Betti Numbers). *At $L = 8$, the kernel dimension is $d = \dim \ker D = b_0 + b_1 + b_2 = 1 + 2 + 1 = 4$.*

2.6.4 What This Changes

Item	Before	After
Derivations of $L = 8$	2 independent	1 theorem, 1 corollary
Derivations of $d = 4$	2 independent	1 theorem, 2 corollaries
Status of $L = 8$	4 consistent derivations	1 theorem, 3 checks
Status of $d = 4$	4 consistent derivations	1 theorem, 3 checks

2.7 The Single Lattice Primitive

The mirror-cancellation theorem of HCSM-10 sharpens the substrate-level axiom structure. Where HCSM-01 lists three substrate-level disclosures, the mirror-cancellation theorem reduces these to a single primitive.

Axiom 2.13 (Lattice Primitive). *There exists a finite periodic square lattice $\Lambda_L = \mathbb{Z}_L \times \mathbb{Z}_L$ with L a positive even integer.*

Every object in the framework, including the substrate size $L = 8$, the 5-point Laplacian, the Hodge complex, the mirror symmetry, the 14-mode multiplet, and the residual 7/32, is a theorem of Axiom 2.13. No physical input, no empirical anchor, and no fitted parameter is used at the substrate level.

2.8 Mirror Cancellation

Definition 2.14 (Half-Period Translation). *The half-period translation is the map*

$$T_{\text{half}}(n_1, n_2) = ((n_1 + L/2) \bmod L, (n_2 + L/2) \bmod L). \quad (17)$$

Lemma 2.15. *T_{half} is an involution on $\{0, \dots, L-1\}^2$.*

Lemma 2.16 (Cosine Shift). *For $n \in \{0, \dots, L-1\}$ with L even,*

$$\cos(2\pi((n + L/2) \bmod L)/L) = -\cos(2\pi n/L). \quad (18)$$

Theorem 2.17 (Mirror Symmetry). *For every momentum pair (n_1, n_2) ,*

$$\lambda(T_{\text{half}}(n_1, n_2)) = 8 - \lambda(n_1, n_2). \quad (19)$$

Consequently, the spectrum of Δ on Λ_L is exactly mirror-symmetric: if λ is an eigenvalue with multiplicity m , then $8 - \lambda$ is also an eigenvalue with multiplicity m .

Corollary 2.18. *Every eigenvalue $\lambda \neq 4$ appears in a mirror pair $(\lambda, 8 - \lambda)$ of equal multiplicity.*

2.9 The 50 + 14 Decomposition and the Residual 7/32

Theorem 2.19 (Self-Paired Eigenvalue). *The unique self-paired eigenvalue is $\lambda = 4$.*

Proof. $\lambda = 8 - \lambda$ implies $2\lambda = 8$, hence $\lambda = 4$. □

Corollary 2.20 (50 + 14 Decomposition). *At $L = 8$, the 64 modes of Λ_8 decompose as*

$$50 \text{ pair modes} + 14 \text{ residual modes at } \lambda = 4. \quad (20)$$

Theorem 2.21 (Residual Ratio 7/32). *For the zero-point convention $F(\lambda) = \lambda/2$ and $L = 8$:*

$$\sum_m F(\lambda_m) = 128, \quad \sum_{m \in \mathcal{R}} F(\lambda_m) = 28, \quad \frac{\sum_{m \in \mathcal{R}} F(\lambda_m)}{\sum_m F(\lambda_m)} = \frac{7}{32}, \quad (21)$$

where $\mathcal{R}$ is the residual set of 14 modes at $\lambda = 4$.

Theorem 2.22 (Convention Independence). *Let $F(\lambda) = \alpha\lambda + \beta$ with $\alpha > 0$ and $F(4) \neq 0$. Then*

$$\frac{\sum_{m \in \mathcal{R}} F(\lambda_m)}{\sum_m F(\lambda_m)} = \frac{7}{32}. \quad (22)$$

If $F(4) = 0$ (equivalently $\beta = -4\alpha$), both numerator and denominator vanish, and the ratio is undefined.

2.10 The TF Operator and the $|\Delta n|$ Grading

Definition 2.23 (Diagonal Flip). The diagonal flip is $F : (n_1, n_2) \mapsto (n_2, n_1)$ on Λ_L .

Definition 2.24 (TF Operator). The TF operator is the composition

$$\text{TF} = T_{\text{half}} \circ F, \quad \text{TF}(n_1, n_2) = ((n_2 + L/2) \bmod L, (n_1 + L/2) \bmod L). \quad (23)$$

Lemma 2.25. $\text{TF}^2 = \text{id}$.

Theorem 2.26 (TF Grading by $|\Delta n|$). *Let $\Delta n = n_2 - n_1$ on Λ_8 . On the 14 modes at $\lambda = 4$:*

- *TF-fixed modes: all modes with $\Delta n = 4 \pmod{8}$. There are 8 such modes: $\{\nu, e, d, b, \mu, c, Z, H\}$.*
- *TF 2-cycle modes: all modes with $\Delta n \neq 4 \pmod{8}$. There are 6 such modes, forming three 2-cycles: $\{u, \tau\}$, $\{s, \text{dark}\}$, $\{W, t\}$.*

Corollary 2.27 (TF-Fixed iff Higher $N/4$). *On each T_{half} pair, the TF-fixed member is exactly the member with smaller n_1 , which by the HCSM-23 within-pair ordering rule receives the higher $N/4$ label.*

Remark 2.28. This corollary derives the HCSM-23 within-pair ordering rule from the TF operator of the substrate. What was stated as a rule in HCSM-23 is now a theorem of Axiom 2.13.

2.11 Klein Four-Orbit Structure

Definition 2.29 (Substrate Involutions). On Λ_8 , three commuting involutions act on the mode labels:

$$F : (n_1, n_2) \mapsto (n_2, n_1), \quad (24)$$

$$\text{TF} : (n_1, n_2) \mapsto ((n_2 + 4) \bmod 8, (n_1 + 4) \bmod 8), \quad (25)$$

$$F \circ \text{TF} : (n_1, n_2) \mapsto ((n_1 + 4) \bmod 8, (n_2 + 4) \bmod 8). \quad (26)$$

All three commute pairwise.

Theorem 2.30 (Klein Four-Orbit). *The four modes $\{u, s, \text{dark}, \tau\}$ carry a faithful action of the Klein four-group $V_4 = \{\text{id}, F, \text{TF}, F \cdot \text{TF}\}$:*

$$F : u \leftrightarrow s, \quad \text{dark} \leftrightarrow \tau, \quad (27)$$

$$\text{TF} : u \leftrightarrow \tau, \quad s \leftrightarrow \text{dark}, \quad (28)$$

$$F \cdot \text{TF} : u \leftrightarrow \text{dark}, \quad s \leftrightarrow \tau. \quad (29)$$

The action is faithful, transitive, and free (no stabilizer).

Corollary 2.31. *The Klein four-orbit $\{u, s, \text{dark}, \tau\}$ is generated by the substrate involutions F and TF . This is the HCSM-27 Klein orbit, now derived from Axiom 2.13.*

3 Level 1: The Hodge Complex

3.1 The Three Form Degrees

On the 2-torus Λ_L , the Hodge complex of differential forms has three graded pieces:

$$\Omega^0 = \mathbb{R}^N, \quad \Omega^1 = \mathbb{R}^{2N}, \quad \Omega^2 = \mathbb{R}^N, \quad (30)$$

with total dimension $N + 2N + N = 4N$. At $N = 64$ this gives 256.

3.2 The Exterior Derivative and the Codifferential

Let $e_1 = (1, 0)$ and $e_2 = (0, 1)$ be the two oriented lattice directions. The exterior derivative acts on 0-forms by

$$(d_0 f)(x, \mu) = f(x + e_\mu) - f(x), \quad \mu \in \{1, 2\}, \quad (31)$$

and on 1-forms by

$$(d_1 \omega)(x) = \omega_1(x) + \omega_2(x + e_1) - \omega_1(x + e_2) - \omega_2(x). \quad (32)$$

These satisfy $d_1 d_0 = 0$ exactly (nilpotency). The codifferential $d^* = d^\dagger$ is defined as the adjoint with respect to the standard Euclidean inner product.

3.3 The Hodge-Dirac Operator

Definition 3.1 (Hodge-Dirac Operator). The Hodge-Dirac operator is $D = d + d^*$, acting on the full Hodge complex $\Omega^0 + \Omega^1 + \Omega^2$. In block form,

$$D = \begin{pmatrix} 0 & d_0^* & 0 \\ d_0 & 0 & d_1^* \\ 0 & d_1 & 0 \end{pmatrix}, \quad (33)$$

a symmetric 256×256 real operator.

Theorem 3.2 (Hodge-Dirac Square). *On the discrete torus Λ , the identity $D^2|_{\Omega^k} = \Delta \cdot \text{id}_{\Omega^k}$ holds exactly on each form degree $k = 0, 1, 2$.*

Proof. On Ω^0 , $D^2 = d_0^* d_0 = \Delta$. On Ω^1 , $D^2 = d_0 d_0^* + d_1^* d_1 = \Delta$ by direct component expansion. On Ω^2 , $D^2 = d_1 d_1^* = \Delta$. Cross terms are removed by nilpotency $d_1 d_0 = 0$. All three identities are verified at machine precision. $\square$

Remark 3.3 (No Species Doubling). The Hodge-Dirac operator on the discrete torus has no species doubling: its square equals the scalar Laplacian on each form degree. The low-energy spectrum is bounded away from zero by $2 \sin(\pi/L)$ except for the four zero modes.

3.4 The $\mathbb{Z}_2$ Grading and the Supercharge Structure

Definition 3.4 (Parity Operator). The parity operator $(-1)^F$ on the Hodge complex is $+1$ on $\Omega^0 + \Omega^2$ and -1 on Ω^1 . It $\mathbb{Z}_2$ -grades the complex into a bosonic sector ($\Omega^0 + \Omega^2$) and a fermionic sector (Ω^1).

Proposition 3.5. $D(-1)^F = -(-1)^F D$.

Theorem 3.6 (Supercharge Structure). *In the $\mathbb{Z}_2$ grading, the Hodge-Dirac operator decomposes as $D = Q + Q^\dagger$, where Q is the off-diagonal block of D mapping the bosonic sector to the fermionic sector, and $Q^\dagger$ is its adjoint. The supercharges satisfy*

$$Q^2 = (Q^\dagger)^2 = 0, \quad \{Q, Q^\dagger\} = D^2 = \Delta, \quad (34)$$

and the Hamiltonian $H = D^2/2 = \Delta/2$ satisfies $\{Q, Q^\dagger\} = 2H$. This is the standard $N = 1$ supersymmetric quantum mechanics algebra.

3.5 The Kernel and the Emergent Dimension

Theorem 3.7 (Kernel Dimension). *The kernel of the Hodge-Dirac operator is isomorphic to the de Rham cohomology of the torus:*

$$\ker D \cong \bigoplus_{k=0}^2 H^k(T^2; \mathbb{R}), \quad (35)$$

with

$$\dim \ker D = b_0 + b_1 + b_2 = 1 + 2 + 1 = 4. \quad (36)$$

Corollary 3.8 (Emergent Dimension). $d = \dim \ker D = 4$.

Theorem 3.9 (Lorentzian Signature). *The Hodge star $*$ restricted to $\ker D$ has eigenvalues $\{+1, -1, +i, -i\}$: the ± 1 pair from the $(H^0 + H^2)$ sector, the $\pm i$ pair from the H^1 sector. The intersection form $Q(\alpha, \beta) = \int_\Lambda \alpha \wedge \beta$ on the even-degree sector and the Hodge inner product $\langle \alpha, \beta \rangle = \int_\Lambda \alpha \wedge * \beta$ on the odd-degree sector combine into a metric*

$$g = Q|_{H^0+H^2} + \langle \cdot, \cdot \rangle|_{H^1} \quad (37)$$

of signature $(3, 1)$ on $\ker D$.

The D_4 action on $\ker D$ decomposes as $\ker D = 2A_1 \oplus E$. The timelike direction is

$$\text{time} = (v_0 - v_2)/\sqrt{2}, \quad (38)$$

where v_0 is the constant 0-form and v_2 the constant 2-form. The spacelike directions are $(v_0 + v_2)/\sqrt{2}$, v_1^A , v_1^B .

3.6 The Spectral Gap and Absence of Species Doubling

Theorem 3.10 (Smallest Nonzero Eigenvalue of D). *On $\Lambda_L = \mathbb{Z}_L \times \mathbb{Z}_L$ with L even, the smallest nonzero eigenvalue of the Hodge-Dirac operator is exactly $\mu_{\min}(L) = 2 \sin(\pi/L)$.*

Corollary 3.11. *For every finite even L , the Hodge-Dirac operator has exactly four zero eigenvalues and the next eigenvalue is bounded away from zero by $2 \sin(\pi/L)$.*

3.7 The Hodge-HCSM Interface: Entropic Flatness

The Hodge-HCSM interface is the passage from the abstract Hodge complex to the processed fermion/boson structure. The mirror-cancellation theorem of Section 2.8 provides the projection onto the 56-dimensional sector V_{56} , and the processor channel of this section provides the map onto the fermion sector.

3.7.1 The 56-Dimensional Hodge Sector

The Hodge complex at the self-paired eigenvalue $\lambda = 4$ is

$$V_{56} = \Omega^0|_{\lambda=4} + \Omega^1|_{\lambda=4} + \Omega^2|_{\lambda=4}, \quad (39)$$

of total dimension $14 + 28 + 14 = 56$. Its D_4 isotypic decomposition is

$$V_{56} = 9A_1 + 5A_2 + 7B_1 + 7B_2 + 14E, \quad (40)$$

with dimension check $9 + 5 + 7 + 7 + 28 = 56$.

3.7.2 The Processor Channel

By HCSM-16 (Definition 1, Lemma 2), the block-diagonal extension map $\text{extend} : \mathbb{C}^{14 \times 14} \rightarrow \mathbb{C}^{56 \times 56}$ is a Lie algebra homomorphism. The 56-dimensional processor is

$$T_{56} = \text{extend}(T) = T + (T \otimes I_2) + T, \quad (41)$$

with $\text{rank}(T_{56}) = 10 + 20 + 10 = 40$ and $\dim \ker(T_{56}) = 16$.

Definition 3.12 (Processor Channel). The processor channel is the completely positive trace-preserving (CPTP) map

$$\Phi_T(\rho) = T_{56}\rho T_{56} + \text{Tr}((I_{56} - T_{56})\rho)|0\rangle\langle 0|_{\text{bos}}, \quad (42)$$

with $|0\rangle_{\text{bos}}$ a fixed reference state on the 16-dimensional bosonic kernel.

3.7.3 Entropic Flatness and the Entropy Reduction

Theorem 3.13 (Entropic Flatness). *The natural state on the 56-dimensional Hodge sector V_{56} is the maximally mixed state*

$$\rho_{\text{in}} = I_{56}/56, \quad (43)$$

obtained as the image of the maximally mixed state on the full Hodge complex $\rho_{\text{full}} = I_{256}/256$ under the mirror-cancellation projection. Its von Neumann entropy is

$$S_{\text{in}} = -\text{Tr}(\rho_{\text{in}} \log \rho_{\text{in}}) = \log 56 \approx 4.0253516907 \text{ nats}. \quad (44)$$

Theorem 3.14 (Processor Channel Output). *The processor channel Φ_T maps the maximally mixed input $\rho_{\text{in}} = I_{56}/56$ to the maximally mixed state on the 40-dimensional image of T_{56} , with von Neumann entropy*

$$S_{\text{out}} = \log 40 \approx 3.6888794541 \text{ nats.} \quad (45)$$

Corollary 3.15 (Entropy Reduction). *The entropy reduction effected by the processor is*

$$\Delta S = S_{\text{in}} - S_{\text{out}} = \log(7/5) \approx 0.3364722366 \text{ nats.} \quad (46)$$

Corollary 3.16 (Landauer Saturation). *Conditional on the HCSM measurement-irreversibility principle (HCSM-21, Theorem 12.1), the entropy reduction $\Delta S = \log(7/5)$ equals the exact deterministic Landauer bound for the processor's information-processing step.*

4 Level 2: The Spectral Midpoint

4.1 The Self-Paired Eigenvalue

The mirror symmetry $\lambda \leftrightarrow 8 - \lambda$ of the scalar Laplacian spectrum has a unique fixed point at $\lambda = 4$. The eigenspace at $\lambda = 4$ has multiplicity $2(L - 1) = 14$ at $L = 8$. The 14 modes are

$$\mathcal{O}_0 = \{(0, 4), (4, 0)\}, \quad (47)$$

$$\mathcal{O}_1 = \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \quad (48)$$

$$\mathcal{O}_2 = \{(2, 2), (2, 6), (6, 2), (6, 6)\}. \quad (49)$$

4.2 The D_4 Orbit Decomposition

Proposition 4.1 (Orbit Stabilizers). *Under the D_4 point group, the 14 modes decompose into the three orbits above of sizes $(2, 8, 4)$. The stabilizer orders of the three orbits are 4, 1, 2 respectively, satisfying the orbit-stabilizer identity*

$$|\mathcal{O}_k| = |D_4|/|\text{Stab}(\mathcal{O}_k)| = 8/|\text{Stab}(\mathcal{O}_k)|. \quad (50)$$

4.3 The 56-Dimensional Hodge Lift

Theorem 4.2 (Fourteen-Fold Lift). *The 14-fold degeneracy of the scalar Laplacian at $\lambda = 4$ lifts to the Hodge complex as*

$$14\Omega^0 + 28\Omega^1 + 14\Omega^2 = 56. \quad (51)$$

4.4 The D_4 Decomposition of the 56-Dimensional Sector

Theorem 4.3 (D_4 Decomposition). *As D_4 representations,*

$$\Omega^0|_{\lambda=4} = 3A_1 + A_2 + 2B_1 + 2B_2 + 3E, \quad (52)$$

$$\Omega^1|_{\lambda=4} = \Omega^0|_{\lambda=4} \otimes E = 3A_1 + 3A_2 + 3B_1 + 3B_2 + 8E, \quad (53)$$

$$\Omega^2|_{\lambda=4} = 3A_1 + A_2 + 2B_1 + 2B_2 + 3E. \quad (54)$$

The full 56-dimensional sector decomposes as $56 = 9A_1 + 5A_2 + 7B_1 + 7B_2 + 14E$.

5 Level 3: The Processor and Its Dynamics

5.1 The Winding Generator

Definition 5.1 (Winding Matrix and Winding Generator). Let $n_1, n_2 \in \mathbb{R}^{14}$ be the coordinate vectors of the 14 modes. The winding matrix is the rank-2 antisymmetric bilinear

$$M_{ij} = n_1^{(i)} n_2^{(j)} - n_1^{(j)} n_2^{(i)}. \quad (55)$$

The winding generator is

$$H_{\text{wind}} = i \sin(\pi M/8), \quad (56)$$

with $\sin$ applied elementwise.

Theorem 5.2 (Characteristic Polynomial). *The characteristic polynomial of H_{wind} is*

$$p(\lambda) = \det(\lambda I_{14} - H_{\text{wind}}) = \lambda^{10}(\lambda^2 - 8)(\lambda^2 - 32). \quad (57)$$

Its spectrum is $\{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\}$.

5.2 The Winding Classes and the Processor

Definition 5.3 (Winding Class Partition). The 14 modes partition into three winding classes:

$$A = \{\nu, e, t\}, \quad |A| = 3, \quad (58)$$

$$B_{\text{wind}} = \{u, d, s, \mu, c, \tau, b\}, \quad |B_{\text{wind}}| = 7, \quad (59)$$

$$C = \{\text{dark}, W, Z, H\}, \quad |C| = 4. \quad (60)$$

The partition is the joint eigenspace decomposition of the n_1 -parity D and the processor T (HCSM-05, Theorem 7.7).

Definition 5.4 (Processor). The processor T is the rank-10 orthogonal projection $T = I - P_C$ with P_C the diagonal projector onto winding class C .

Proposition 5.5 (Fermion/Boson Split). $T = I - P_C = P_A + P_{B_{\text{wind}}}$, with $\text{im}(T) = F = A \cup B_{\text{wind}}$ and $\ker(T) = C = B_{\text{bos}}$. The fermion and boson sector dimensions are $|F| = d(d+1)/2 = 10$ and $|B_{\text{bos}}| = d = 4$ (HCSM-13, HCSM-19). The internal ratios are $(|F| - |B_{\text{bos}}|)/|B_{\text{bos}}| = (d-1)/2 = 3/2$ and $|F|/|B_{\text{bos}}| = (d+1)/2 = 5/2$.

5.2.1 Chirality-Breaking Localization

Theorem 5.6 (Chirality-Breaking Localization). *The commutator of the processor with the chiral reflection,*

$$C_{rs} = T \cdot P_{rs} - P_{rs} \cdot T, \quad (61)$$

is supported exactly on the two-element set $\{\text{dark}, \tau\}$. On this two-dimensional plane, the commutator is a fourth-order rotation generator:

$$C_{rs}|_{\text{dark}, \tau} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad (62)$$

with $C_{rs}^2|_{\text{dark}, \tau} = -P_{\text{dark}, \tau}$ and $C_{rs}^4|_{\text{dark}, \tau} = +P_{\text{dark}, \tau}$. On the other twelve modes, T and P_{rs} commute.

5.2.2 The Observer Pair

Proposition 5.7 (Observer Pair). *The chiral reflection rs fixes exactly the two modes $W = (2, 2)$ and $t = (6, 6)$. These form the observer pair of HCSM-26.*

5.3 The Kernel-Diagonal Theorem

Theorem 5.8 (Kernel-Diagonal). *Let $P_{\ker}$ be the orthogonal projector onto $\ker(H_{\text{wind}})$. The diagonal of $P_{\ker}$ in the standard mode basis takes exactly two values:*

$$\text{diag}(P_{\ker})_i = \begin{cases} 2/d = 1/2, & i \in \mathcal{O}_0, \\ (d-1)/d = 3/4, & i \in \mathcal{O}_1 \cup \mathcal{O}_2. \end{cases} \quad (63)$$

Corollary 5.9 (Three Coefficients from One Diagonal). *With $P_{\ker}$ as above,*

$$c = \frac{d+1}{\text{Tr}(P_{\ker}|_{\mathcal{O}_1 \cup \mathcal{O}_2})} = \frac{5}{9}, \quad (64)$$

$$\frac{\text{diag}(P_{\ker})|_{\mathcal{O}_1 \cup \mathcal{O}_2}}{\text{diag}(P_{\ker})|_{\mathcal{O}_0}} = \frac{d-1}{2} = \frac{3}{2}, \quad (65)$$

$$\dim \ker(H_{\text{wind}}) = 1 + (d-1)^2 = 10. \quad (66)$$

5.3.1 Kernel- T Decomposition

Theorem 5.10 (Kernel- T Decomposition). *The kernel of H_{wind} decomposes under the processor T as*

$$\dim(\ker H_{\text{wind}} \cap \text{im } T) = 6, \quad \dim(\ker H_{\text{wind}} \cap \ker T) = 1. \quad (67)$$

The remaining 3 dimensions of $\ker H_{\text{wind}}$ form a mixed subspace with support on both F and C . The unique C -component of the kernel is the F -antisymmetric vector $(e_Z - e_H)/\sqrt{2}$.

5.4 The Five-Layer Loop

The five-layer composition acts diagonally on the 14-mode multiplet:

$$M_{5\text{layer}} = T \cdot L_{\text{EFT}} \cdot L_{\text{Core}} \cdot L_{\text{DME}} \cdot L_{\text{FDLM}}. \quad (68)$$

Theorem 5.11 (Loop-Pass Duration). *The loop-pass duration is selected by the deterministic Lander bound:*

$$T_{\text{pass}} = T_{\text{DME}} = \frac{1}{2\gamma} = 0.7114000035169658 \dots \quad (69)$$

Theorem 5.12 (Exact Ratio). $|M_A|/|M_B| = 3 = d - 1$.

5.4.1 Family-Block Eigenvalues

Theorem 5.13 (Family-Block Eigenvalues). *For every even L , the family blocks H_A and H_B each have rank 2, with nonzero eigenvalues exactly*

$$\text{spec}_{\neq 0}(H_A) = \text{spec}_{\neq 0}(H_B) = \{+L/2, -L/2\}. \quad (70)$$

5.4.2 Four Kernel Constants

Theorem 5.14 (Kernel Constants at $L = 8$). *At $L = 8$, the kernel of $H_{\text{wind}}(8)$ contains the four parity-block indicators*

$$1_{A_{\text{odd}}}, \quad 1_{A_{\text{even}}}, \quad 1_{B_{\text{odd}}}, \quad 1_{B_{\text{even}}}. \quad (71)$$

Their supports are pairwise disjoint, so they span a 4-dimensional subspace of $\ker H_{\text{wind}}(8)$. The fifth indicator 1_{AB} is not in the kernel: $\|H_{\text{wind}}(8)1_{AB}\| = 4$.

5.4.3 Rank Formula

Theorem 5.15 (Rank Formula). *For every even $L \geq 4$,*

$$\text{rank } H_{\text{wind}}(L) = \begin{cases} L, & L \equiv 2 \pmod{4}, \\ 4\lceil L/8 \rceil, & L \equiv 0 \pmod{4}. \end{cases} \quad (72)$$

5.5 The Loop Operator and Its Commutant

The loop operator M_{loop} is the diagonal class operator of HCSM-17 Definition 2.1, distinct from the five-layer composition $M_{5\text{layer}}$.

Definition 5.16 (Loop Operator). The loop operator on the 14-mode multiplet is the diagonal operator

$$M_{\text{loop}} = M_A P_A + M_B P_B + M_C P_C, \quad (73)$$

with M_A, M_B, M_C the complex loop eigenvalues of HCSM-14 and HCSM-17, and $|M_A|/|M_B| = 3$ exactly.

Definition 5.17 (Loop Commutant). The loop commutant is the intersection

$$\mathfrak{g}_{\text{loop}} = \mathfrak{g}_{\text{SM}} \cap \text{Comm}(M_{\text{loop}}) \subset \mathfrak{g}_{\text{SM}}. \quad (74)$$

Theorem 5.18 (Loop Commutant Dimension). *The loop commutant $\mathfrak{g}_{\text{loop}}$ is a three-dimensional real, Hermitian, abelian subalgebra of $\mathfrak{g}_{\text{SM}}$.*

Theorem 5.19 (Isotypic Projector Degeneracy). *Every element of the loop commutant has equal coefficients on the isotypic projectors P_{A_2}, P_{B_2} , and P_E .*

Theorem 5.20 (No Root Generators). *The loop commutant has zero overlap with the root generators $\lambda^1, \lambda^2, \lambda^4, \lambda^5, \lambda^6, \lambda^7$ of $\mathfrak{su}(3)_c$ and σ^1, σ^2 of $\mathfrak{su}(2)_L$. Only the Cartan generators $\lambda^3, \lambda^8, \sigma^3$ of $\mathfrak{g}_{\text{SM}}$ have nonzero overlap with the loop commutant.*

5.6 The DNLS Hartree Hamiltonian and Its Commutant

Definition 5.21 (DNLS Hartree Kernel). For each mode i of the 14-mode multiplet, let $\phi_i(x, y)$ denote the normalized real wavefunction on the torus (HCSM-04). The Hartree kernel is

$$K_{ab} = \sum_{x \in \Lambda} |\phi_a(x)|^2 |\phi_b(x)|^2, \quad a, b = 1, \dots, 14. \quad (75)$$

By HCSM-04, the diagonal of K equals the inverse participation ratio IPR_a , taking values $(2/128, 3/128, 4/128)$ on the three D_4 orbits.

Definition 5.22 (DNLS Hartree Hamiltonian). The DNLS Hartree Hamiltonian on the 14-mode multiplet is

$$H_{\text{DNLS}} = 4I_{14} + UK, \quad (76)$$

with $U = B_{\text{mass}} = (d-1)kL^2/(2N^2) = 0.5411967342\dots$ the mass-map rate (HCSM-19, HCSM-33).

Definition 5.23 (DNLS Commutant). The DNLS commutant is

$$\mathfrak{g}_{\text{DNLS}} = \mathfrak{g}_{\text{SM}} \cap \text{Comm}(H_{\text{DNLS}}). \quad (77)$$

Theorem 5.24 (DNLS Commutant Dimension). *The DNLS commutant $\mathfrak{g}_{\text{DNLS}}$ is a ten-dimensional real, Hermitian subalgebra of $\mathfrak{g}_{\text{SM}}$, isomorphic to $\mathfrak{su}(2)_L + \mathfrak{u}(1)^7$.*

5.6.1 Doubled-Momentum Pairing

Lemma 5.25 (Doubled-Momentum Pairing). *Let $U_{B_1} = [v_1^{(B_1)}, v_2^{(B_1)}] \in \mathbb{R}^{14 \times 2}$ be the explicit orthonormal basis of the B_1 isotypic sector constructed in Appendix A. The two basis vectors pair modes of the multiplet such that for each pair $\{m, m'\}$ in the support of $v_1^{(B_1)}$ and $v_2^{(B_1)}$, the doubled momenta agree:*

$$(2n_1^{(m)}, 2n_2^{(m)}) = (2n_1^{(m')}, 2n_2^{(m')}) \pmod{8}, \quad (78)$$

and the mode densities satisfy

$$|\phi_m(x)|^2 = |\phi_{m'}(x)|^2, \quad (79)$$

with the sign opposite between the two B_1 basis vectors.

Theorem 5.26 (Doubled-Momentum Pairing Theorem). *The B_1 -compressed Hartree kernel vanishes identically:*

$$K_{B_1} = U_{B_1}^T K U_{B_1} = 0 \in \mathbb{R}^{2 \times 2}. \quad (80)$$

Corollary 5.27 ($\mathfrak{su}(2)_L$ Preservation). *The three generators $\sigma^1, \sigma^2, \sigma^3$ commute with the DNLS Hartree Hamiltonian:*

$$[\sigma^i, K] = 0, \quad [\sigma^i, H_{\text{DNLS}}] = 0, \quad i = 1, 2, 3. \quad (81)$$

Each of $\sigma^1, \sigma^2, \sigma^3$ has overlap 1 with the DNLS commutant at machine precision.

5.6.2 $\mathfrak{su}(3)_c$ Breaking

Theorem 5.28 ($\mathfrak{su}(3)_c$ Breaking). *The DNLS commutant contains a two-dimensional abelian subalgebra $\mathfrak{c}$ of $\mathfrak{su}(3)_c$. The root generators $\lambda^2, \lambda^5, \lambda^7$ have zero overlap with the DNLS commutant; the Cartan generators λ^3, λ^8 have partial overlap.*

5.7 The Zero State

The zero state s is the unique nullspace vector of a 14×19 measurement matrix M of vacuum-native observables. The matrix has rank 13, nullity 1.

Theorem 5.29 (Zero-State Structure). *The zero state satisfies $s(rs(m)) = -s(m)$ for all modes m , where rs is the diagonal reflection $(n_1, n_2) \mapsto (n_2, n_1)$. The two rs -fixed modes $W = (2, 2)$ and $t = (6, 6)$ have $s(W) = s(t) = 0$ exactly. The six independent amplitudes satisfy $a_3 = 2a_4$ and $a_5 = (2 - \sqrt{2})a_4$. Weight fractions: 3.3% on $\mathcal{O}_0$, 86.5% on $\mathcal{O}_1$, 10.2% on $\mathcal{O}_2$.*

5.8 The 5+9 Split

The 5+9 split of the 14-mode multiplet (HCSM-11) refines the processor's action. The classification is defined by the residual

$$\delta = N/4 - \text{base}, \quad (82)$$

where base is a reference depth in the entropy well.

Definition 5.30 (Substrate Eigenmode and Mesh Resonance). A mode is a Substrate Eigenmode if $\delta = 0$; it is a Mesh Resonance if $\delta \neq 0$.

Theorem 5.31 (Derived Base Formula). *For each mode at $\lambda = 4$, let (k, node) be its T_{half} pair label. The residual $\delta = N/4 - \text{base}$ is given by*

$$\delta_{\text{high}} = (d-1)(2-k) + (d-1)^3 K[\text{node} = AB], \quad (83)$$

$$\delta_{\text{low}} = \text{frac}(N/4_{\text{low}}) + 2(d-1)K[\text{node} = AB], \quad (84)$$

where $\text{frac}(x) = x - \lfloor x \rfloor$ and $K[\cdot]$ is the indicator function. The base is $\text{base} = N/4 - \delta$.

Theorem 5.32 (5+9 Split). *Of the 14 modes at $\lambda = 4$, exactly 5 are Substrate Eigenmodes and exactly 9 are Mesh Resonances. The 5 Substrate Eigenmodes are the dark photon, the tau lepton, the W boson, the Z boson, and the Higgs boson. The 9 Mesh Resonances are the remaining 9 fermions.*

Theorem 5.33 (Shell Formula). *The shell of each mode is determined by the $N/4$ value through the step function*

$$\text{shell}(N/4) = \begin{cases} 0, & N/4 \geq 50, \\ 1, & 18 \leq N/4 < 50, \\ 1.5, & 12 \leq N/4 < 18, \\ 2, & 6 \leq N/4 < 12, \\ 3, & N/4 < 6. \end{cases} \quad (85)$$

The thresholds are framework constants: $\{2(d-1), d(d-1), 2(d-1)^2, N-2(L-1)\} = \{6, 12, 18, 50\}$.

Theorem 5.34 (W/Z Base Formula). *The base values of the W and Z bosons are $\text{base}(W) = \text{base}(Z) = d/2 = 2$.*

5.9 The Four-Layer Architecture

Theorem 5.35 (Layer Independence). *The 14-mode multiplet carries four independent structural layers:*

1. **Winding layer:** the kernel projector P_{ker} of H_{wind} , producing $c = 5/9$, $(d-1)/2 = 3/2$, and kernel dimension 10.
2. **Hartree layer:** the mode-density kernel K , producing the eigenvalue $7/32$ and the coefficient $26 = 2^{d+1} - (|F| - |B_{\text{bos}}|)$.
3. **Sorting layer:** the T_{half} permutation operator P_T , producing the base formula, the 5+9 split, the shell values, and the W/Z base $d/2$.
4. **Processing layer:** the shape parameters r_a and the coefficients $\{7.88, 4.23\}$ of $m_f(r)$, which are orthogonal to layers 1, 2, and 3.

Layers 1, 2, and 3 are HCSM-native and mutually compatible: $[P_T, K] = 0$, $[P_{\text{ker}}, F] = 0$, $[P_{\text{ker}}, D] = 0$, and K commutes with F and D . Layer 4 is orthogonal to all three.

5.10 The Mirror Residual as a Hartree Eigenvalue

Theorem 5.36 (Mirror Residual equals Hartree Eigenvalue). *The top eigenvalue of the Hartree kernel K equals the mirror-cancellation residual of Section 2.9:*

$$\lambda_{\max}(K) = 7/32 = (2d - 1)/2^{d+1}. \quad (86)$$

The corresponding eigenvector is the uniform superposition of all 14 modes.

Theorem 5.37 (T_{half} Commutes with the Hartree Kernel). *The T_{half} permutation operator P_T and the Hartree kernel K commute:*

$$[P_T, K] = 0. \quad (87)$$

Theorem 5.38 (Coefficient 26). *The HCSM-30 processing function coefficient 26 satisfies*

$$26 = 2^{d+1} - (|F| - |B_{\text{bos}}|) = 32 - 6. \quad (88)$$

6 Level 4: Output Structures

6.1 The Charge Functional

The electric charge in units of $e/3$ is given by the functional

$$q_3 = F(n_1, \sigma), \quad \sigma = \text{sign}(s), \quad (89)$$

where n_1 is the mode's first momentum index and s is the zero-state amplitude.

Theorem 6.1 (Charge Bijection). *The 14 modes have 14 pairwise distinct (n_1, σ) tuples. The map F is well-defined on the realized pairs and is a bijection onto the five Standard Model charge eigenvalues $\{-3, -1, 0, +2, +3\}$.*

Theorem 6.2 (σ -Line Sums). *The charge sums over the three σ -lines are*

$$\sum_{\sigma=-1} q_3 = 0, \quad \sum_{\sigma=0} q_3 = +5, \quad \sum_{\sigma=+1} q_3 = -8. \quad (90)$$

The total vacuum charge is $\sum_i q_{3i} = -3 = -1e$.

6.2 The Rigidity Class of the Charge Assignment

Theorem 6.3 (Charge Rigidity Class). *Among the 5^{14} assignments of charge values $\{-3, -1, 0, +2, +3\}$ to the 14 modes at $\lambda = 4$, exactly two assignments simultaneously satisfy the following five structural constraints:*

1. **(C1) SM multiset.** *The multiset of charges is $\{-3, -1, 3, 0, 4, +2, 3, +3, 1\}$.*
2. **(C2) σ -line sums.** *The charge sums over the three zero-state sign lines are 0, $d + 1 = 5$, and $-2d = -8$.*
3. **(C3) F-pair sum rules.** *Within each $|\Delta n|$ shell, the pair charge sum is rank- d (for $|\Delta n| = 4$) or $(d + 1) - d$ rank (for $|\Delta n| = 2$).*
4. **(C4) $n_1 + n_2 = 4$ line.** *The five modes on the line $n_1 + n_2 = 4$ carry the five charge values exactly once each.*

5. **(C5) Processor bookkeeping.** *The image-of- T charge sum is -2 and the kernel-of- T charge sum is $+1$.*

Theorem 6.4 (Total Charge Rigidity). *Both solutions have total charge $\sum_i q_{3i} = -3 = -1e$. Consequently, every assignment satisfying the five structural constraints has total charge $-1e$:*

$$\sum_i q_{3i} = -(d-1) = -1e. \quad (91)$$

The total vacuum charge is a rigidity-class invariant.

6.3 The Mass Map

The mass map is

$$m_i = A_{\text{mass}} \exp(-B_{\text{mass}}(N/4)_i), \quad (92)$$

with

$$A_{\text{mass}} = \frac{(2d-1)N}{2} = 224 \text{ GeV}, \quad B_{\text{mass}} = \frac{(d-1)kL^2}{2N^2} = 0.5411967342\dots \quad (93)$$

The $N/4$ labels are derived from framework constants (HCSM-23 Rev. 8, Theorem 4.1):

$$(N/4)_i = (60, 24, 21, 20, 14, 14, 12, 9.5, 9, 7.5, 2, 2, 1, 0.5). \quad (94)$$

The map reproduces the twelve charged fermion masses and the W, Z, H, and t masses at RMS 0.0449 dex across thirteen orders of magnitude. The coefficient $(d-1)/2$ in B_{mass} is derived in HCSM-19 by two independent routes (the processor F/B split and the DNLS Hartree orbit trace). The mass map itself is derived in HCSM-33 Rev. 2.

The geometric-mean identity is

$$m_\nu M_P = A_{\text{mass}} v_{\text{EW}} \exp(2(d-1)). \quad (95)$$

6.4 The Midpoint Sorting Theorem

The identification of the 14 modes with one Standard Model generation is a theorem (HCSM-23 Rev. 8). The sorting uses the T_{half} involution $(n_1, n_2) \mapsto (n_1+4, n_2+4)$, the pair index $k = \min(n_1)$, the parity-block node from the sign of the signed coordinate v of the smaller- n_1 member, the (S, D) table of HCSM-23 Rev. 8 (now a corollary of the $N/4$ derivation), and the within-pair ordering rule.

The central new result of the Eighth Revision: the fourteen $N/4$ labels are elementary functions of the framework integers $\{d, N, L, H\} = \{4, 64, 8, 16\}$:

$$N/4(\nu) = N - d = 60, \quad N/4(e) = d! = 24, \quad (96)$$

$$N/4(u) = (d-1)(2d-1) = 21, \quad N/4(d) = N/4 + d = 20, \quad (97)$$

$$N/4(s) = N/4(\mu) = 2(L-1) = 14, \quad N/4(\text{dark}) = (d-1)d = 12, \quad (98)$$

$$N/4(c) = (d^2 + 3)/2 = 9.5, \quad N/4(\tau) = (d-1)^2 = 9, \quad (99)$$

$$N/4(b) = (d^2 - 1)/2 = 7.5, \quad N/4(W) = N/4(Z) = d/2 = 2, \quad (100)$$

$$N/4(H) = 1, \quad N/4(t) = 2/d = 0.5. \quad (101)$$

7 Level 5: Emergent Physics

7.1 Gauge Algebra from the D_4 Commutant

The gauge algebra $\mathfrak{su}(3) + \mathfrak{su}(2) + \mathfrak{u}(1)^5$ is derived as a Lie subalgebra of the D_4 commutant on the 14-mode multiplet (HCSM-06). The A_1 isotypic block is uniquely identified with physical color by a dimension-and-invariance theorem (HCSM-06, Theorem 5.4). The B_1 block is identified with physical weak isospin up to the outer-automorphism labeling convention (HCSM-06, Theorem 5.10).

7.2 Hypercharge Quantization

The hypercharge functional is a specific element of the 15-dimensional sector-coarse space (HCSM-06, Theorem 6.3; HCSM-07). The A_1 coefficients are $(0, -1/6, -1/2)$ and the B_1 coefficients are $(0, +1/2, +1/2)$, uniquely determined by the Standard Model content and the sector assignment (HCSM-07, Theorems 5.1 and 6.1). The hypercharge values are derived from the content structure and the anomaly cancellation conditions (HCSM-07, Theorem 8.1). The content structure is derived from the 14-mode charges, the D_4 isotypic multiplicities, and the Hodge-lift chirality structure (HCSM-07, Theorem 8.3). All values lie in $(1/6)\mathbb{Z}$.

The $\mathfrak{u}(1)_Y$ origin is derived from the conserved hypercharge functional $q_3 = F(n_1, \sigma)$ on $\Omega^0|_{\lambda=4}$, whose generator $\hat{Y}$ commutes with the processor T and with the orientation $\mathbb{Z}_2 F$ (HCSM-44).

7.3 Flavor: The CKM Matrix

The Cabibbo angle satisfies $\sin \theta_{12} = \pi/14 = 0.2243995\dots$, matching observation to 0.044% (HCSM-34, Theorem 1). The winding amplitude $a = 3/5$ is topological: $a = (|F| - |B_{\text{bos}}|)/|F| = 6/10$. The Wolfenstein parameters are $\lambda = \pi/14$, $A = (d/(d+1))(1 + \delta_{\text{edge}})$, and $\sqrt{\bar{\rho}^2 + \bar{\eta}^2} = (d+1)/(d(d-1)) = 5/12$ (HCSM-34, Theorem 3). The full CKM hierarchy is derived in HCSM-34, Theorems 5 and 9.

The nine CKM elements partition into three $\mathbb{Z}_3$ -cyclic classes, each satisfying a daughter relation: the smallest element equals the product of the other two times a coefficient c_k . The three coefficients are

$$c_0 = 1, \quad c_1 = \frac{N}{N + (d-1)^d} = \frac{64}{145}, \quad c_2 = w_d \left(1 + \frac{m_u}{m_c} \cdot \frac{N-14}{N-13} \right), \quad (102)$$

with $w_d = (2d-1)/(3d-2) = 7/10$ (HCSM-35, Theorem 6.2). The factor $(N-14)/(N-13) = 50/51$ is the residual-bath coupling strength of the Hodge complex: the ratio of cancelling paired modes to cancelling plus observer modes.

The framework predicts $|V_{td}| = 0.0061 \pm 0.0001$, testable by Belle II and LHCb in 2026–2028.

7.4 Flavor: The PMNS Matrix

The neutrino $N/4$ spectrum is $(60, 60, 58)$, derived from the four spin structures of the torus with Dirac indices $\pm 1/4$ (HCSM-36, Theorem 4.1). The PMNS magnitudes match at RMS 0.035 (HCSM-36, Theorem 6.1). The mixing angles are $\theta_{12} = 34.2^\circ$, $\theta_{23} = 47.9^\circ$, $\theta_{13} = 8.5^\circ$; the Dirac CP phase is $\delta_{CP} \approx 197^\circ$. The generation phase is

$$\phi_{\text{gen}} = \frac{|F| - |B_{\text{bos}}|}{|F|} \cdot \pi = \frac{3}{5}\pi = 0.6\pi \quad (103)$$

(HCSM-37, Theorem 3.1), unifying with the CKM winding-phase amplitude $a = 3/5$.

The CKM/PMNS asymmetry is explained by the different $N/4$ spectra of the quark and lepton sectors: separated quark labels (21, 9.5, 0.5 up-type; 20, 14, 7.5 down-type) give small mixing; nearly degenerate neutrino labels (60, 60, 58) give large mixing.

7.5 Higgs Sector

The Higgs doublet is the complexification of the B_1 isotypic of $\Omega^0|_{\lambda=4}$ (HCSM-40, Theorem 3.2). The physical weak direction is fixed by the vacuum zero state s . The DNLS Hartree Hamiltonian preserves $\mathfrak{su}(2)_L$ exactly for every interaction strength (HCSM-40, Theorem 5.1).

The two-loop anomaly closes by a geometric mechanism: $\{\gamma_5, D\} = 0$ forces $\text{Tr}(\gamma_5 f(D)) = 0$ for every even regulator, independent of Adler-Bardeen (HCSM-40, Theorem 6.3).

The top-Yukawa coefficient is $4/3 = |\mathcal{O}_2|/(d-1)$, where $\mathcal{O}_2$ is the unique orbit of the 14-mode multiplet fixed entirely by the mirror map $M : (n_1, n_2) \mapsto (4-n_1, 4-n_2)$ (HCSM-40, Theorem 7.2). The formula $y_t^2 = 1 - (4/3)\delta_{\text{edge}}$ predicts $m_t = 172.688398$ GeV, matching PDG to 0.0053σ .

The tree-level quartic coupling is

$$\lambda_{\text{tree}} = \frac{\ln(kL)}{(d-1)\pi^2}, \quad (104)$$

matching the observed m_H to 9.5×10^{-7} relative. The bridge identity

$$\lambda_{\text{tree}} = \frac{\gamma^2}{d} \left(1 - \frac{\delta_{\text{edge}}}{2(d-1)} \right) \quad (105)$$

holds to 8.4×10^{-8} (HCSM-40, Theorem 9.1).

The electroweak scale is the single dimensionful anchor:

$$v_{\text{EW}} = M_P \exp(-kL_{\text{phys}}). \quad (106)$$

7.6 Gravity

The graviton is the self-paired $B_1 + B_2$ curvature mode of $\text{Sym}^2(\Omega^1|_{\lambda=4})$, consisting of $54B_1 + 54B_2 = 108$ modes (HCSM-45, HCSM-46). It is exactly massless at all orders by D_4 Schur orthogonality: any trace-coupled mass operator transforms in A_1 , while the graviton transforms in $B_1 + B_2$; the irreps are distinct, so $\langle A_1, B_1 + B_2 \rangle_{D_4} = 0$ (HCSM-46, Theorem 4).

The symmetric square decomposes as

$$\text{Sym}^2(\Omega^1|_{\lambda=4}) = 60A_1 + 46A_2 + 54B_1 + 54B_2 + 96E, \quad (107)$$

of total dimension 406. The graviton polarization space is $V = B_1 + B_2$, of dimension 2, with character $(2, 2, -2, 0, 0)$. The B_1 component is the $h_{xx} - h_{yy}$ polarization; the B_2 component is the h_{xy} polarization.

The graviton self-coupling is determined by the D_4 -invariant content of the graviton vertex spaces. The pure-polynomial A_1 counts are $(0, 2, 0, 3)$ at orders 1–4, forbidding pure-polynomial kinetic and cubic self-couplings. The derivative-included counts with two derivatives are $(5, 12, 11)$ at orders 2, 3, 4, establishing the representation-theoretic content required for the kinetic, cubic, and quartic vertices (HCSM-47).

The Newton constant is

$$N = \frac{(d+1)^2}{8\pi M_P^2} = \frac{25}{8\pi M_P^2}, \quad (108)$$

derived from first principles in HCSM-48 (the bilinear trace $25 = (d+1)^2$ from the hidden B_1 isotypic sector) and HCSM-49 (the $8\pi = 2\pi \cdot \dim \ker D = 2\pi \cdot 4$ factor from the Faddeev-Popov measure). The hidden B_1 sector has dimension 5, decomposing as $3_{\Omega^1} + 2_{\Omega^2} = (d-1) + (d/2) = d+1$. The Faddeev-Popov cancellation identity $\det(\Delta - 4I)|_{B_1 \subset \Omega^0, \text{nonzero}} = 2^5 = 32$ selects B_1 uniquely among the five D_4 irreps.

7.7 Absence of a Light Dark Photon

Theorem 7.1 (Absence of a Light Dark Photon). *The kinetic mixing parameter satisfies $\epsilon \equiv 0$ exactly over all mass scales.*

Proof. The HCSM gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$ contains no hidden Abelian gauge factor that couples to electrically charged matter. The hidden sector content $5 + 6 + 31$ (HCSM-38) is $U(1)_Y$ -neutral (HCSM-39, Disclosed Choice 2). No state is charged under both the visible $\mathfrak{u}(1)_{\text{EM}}$ and any hidden Abelian factor. Therefore no tree-level kinetic mixing operator $F_{\mu\nu}F'^{\mu\nu}$ can be written, and no one-loop kinetic mixing is generated. The parameter ϵ does not exist as a degree of freedom in the HCSM effective theory. $\square$

The prediction is falsifiable by every major dark photon search: SHiP, FASER2, MATHUSLA, CODEX-b, LDMX, NA64, Belle II, and HL-LHC. Any confirmed dark photon with $\epsilon > 10^{-12}$ at any mass below the electroweak scale falsifies the framework.

7.8 Spin Chirality $\mathbb{Z}_2$ Grading

The 56-dimensional Hodge lift at $\lambda = 4$ carries three commuting $\mathbb{Z}_2$ operators:

Definition 7.2 (Chirality Operator γ_5). The Hodge form-parity $\gamma_5 = P_{\Omega^0} - P_{\Omega^1} + P_{\Omega^2}$, acting as $+1$ on even forms ($\Omega^0 \oplus \Omega^2$) and -1 on odd forms (Ω^1). It satisfies $\gamma_5^2 = I$ and $\{\gamma_5, D\} = 0$.

Definition 7.3 (Spin-Structure Parity χ_{spin}). The spin-structure parity χ_{spin} distinguishes the two even spin structures (NS, NS) and (R, R) from the two odd spin structures (NS, R) and (R, NS), with eigenvalue $+1$ on the even pair and -1 on the odd pair.

Definition 7.4 (Orientation $\mathbb{Z}_2$). The orientation flip $F : (n_1, n_2) \mapsto (n_2, n_1)$, which preserves each D_4 orbit, commutes with Δ , and anticommutes with the winding generator: $\{F, H_{\text{wind}}\} = 0$.

Theorem 7.5 ($\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ Structure). *The three operators γ_5 , χ_{spin} , and F pairwise commute on the 56-dimensional Hodge lift at $\lambda = 4$:*

$$[\gamma_5, \chi_{\text{spin}}] = 0, \quad [\gamma_5, F] = 0, \quad [\chi_{\text{spin}}, F] = 0. \quad (109)$$

Each squares to the identity. Together they generate a group isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ acting on the 56-dimensional Hodge lift.

The three $\mathbb{Z}_2$ structures combine to give the physical spin-statistics, chirality, and CP properties of the Standard Model at the substrate level: γ_5 gives spin-statistics (even forms bosonic, odd forms fermionic); χ_{spin} gives chirality (even structures left-handed, odd structures right-handed); F gives CP (orientation flip).

The 56-dimensional Hodge lift splits into a bosonic sector $\mathcal{B} = (\Omega^0 \oplus \Omega^2)|_{\lambda=4}$ of dimension 28 and a fermionic sector $\mathcal{F} = \Omega^1|_{\lambda=4}$ of dimension 28.

7.9 Strong CP from the Orientation $\mathbb{Z}_2$

Theorem 7.6 (Strong CP). $\theta_{\text{QCD}} = \theta_{\text{eff}} = 0$ exactly.

Proof. On the 14-mode multiplet at $\lambda = 4$, $[F, \Delta] = 0$ and $\{F, H_{\text{wind}}\} = 0$ exactly (HCSM-43, Theorem 4.1). The winding matrix M is antisymmetric under F ; the odd function $\sin(\pi M/8)$ preserves the antisymmetry. The substrate vacuum lies in an F -eigensector, so every F -odd observable has

vanishing vacuum expectation value. Since H_{wind} is F -odd and is identified with the substrate realization of $F_{\mu\nu}\tilde{F}^{\mu\nu}$, the HCSM vacuum forces $\theta_{\text{QCD}} = 0$ exactly. The HCSM mass map produces positive diagonal masses, so $\arg \det(M_u M_d) = 0$ and therefore $\theta_{\text{eff}} = \theta_{\text{QCD}} + \arg \det(M_u M_d) = 0$ exactly. No axion, no Peccei-Quinn symmetry, no Nelson-Barr mechanism, and no new degrees of freedom are required. $\square$

7.10 Proca Sector and $u(1)_Y$ Origin

Definition 7.7 (Hypercharge Generator $\hat{Y}$). On the visible sector $\Omega^0|_{\lambda=4}$, the hypercharge generator is

$$\hat{Y} = \sum_{i \in \Omega^0|_{\lambda=4}} Y(m_i) |m_i\rangle \langle m_i|, \quad (110)$$

where $Y(m_i)$ is the hypercharge value assigned to the mode m_i by the HCSM-07 hypercharge functional.

The $u(1)_Y$ gauge symmetry is the continuous symmetry generated by $\hat{Y}$. Its generator satisfies $[\hat{Y}, T] = 0$ and $[\hat{Y}, F] = 0$ (HCSM-44, Theorem 3.2).

The Proca Lagrangian is

$$\mathcal{L}_{\text{Proca}} = -\frac{1}{4} W_{\mu\nu}^a W^{a\mu\nu} - \frac{1}{4} B_{\mu\nu} B^{\mu\nu} + \frac{1}{2} m_W^2 W_\mu^+ W^{-\mu} + \frac{1}{2} m_Z^2 Z_\mu Z^\mu, \quad (111)$$

with

$$m_W = \frac{g^{\text{vEW}}}{2} = 80.35 \text{ GeV}, \quad m_Z = \frac{\sqrt{g^2 + g'^2} v_{\text{EW}}}{2} = 91.63 \text{ GeV}, \quad \frac{m_W}{m_Z} = \cos \theta_W = 0.8769. \quad (112)$$

The mass ratio matches PDG 0.8814 at 0.5%. Tree-level $\rho = 1$ exactly.

The photon is the massless eigenstate $A_\mu = \sin \theta_W W_\mu^3 + \cos \theta_W B_\mu$ of the $SU(2)_L \times U(1)_Y$ mass matrix. Transversality $\partial^\mu A_\mu = 0$ is derived from the Higgs mechanism, and the Goldstone boson is eaten.

7.11 Graviton Masslessness

Theorem 7.8 (Graviton Exact Masslessness). *The graviton is exactly massless at all orders in perturbation theory and at all orders in $1/N$: $m_{\text{graviton}} = 0$.*

Proof. Any mass operator on the $\lambda = 4$ sector that couples through the trace ϵT_μ^μ (with $\epsilon = D - 4$ a Lorentz scalar) transforms in the A_1 isotypic representation of D_4 . The graviton sector is $V_{\text{grav}} = 54B_1 + 54B_2$, which lies entirely in the $B_1 + B_2$ isotypic sector. The irreps A_1 and B_1, B_2 are distinct irreducible representations of D_4 . By Schur's lemma, any D_4 -invariant operator is block-diagonal with respect to the isotypic decomposition, so $\langle A_1, B_1 + B_2 \rangle_{D_4} = 0$. The mass operator therefore annihilates every mode in $B_1 + B_2$, and the graviton mass vanishes. The argument is exact and does not rely on smallness of any coupling, on a perturbative expansion, or on a finite- N suppression. $\square$

7.12 Faddeev-Popov G_N Normalization

Theorem 7.9 (Faddeev-Popov G_N Normalization). *The kinetic-term coefficient in the HCSM Newton constant is exactly*

$$C_{\text{kin}} = 25 \cdot \frac{2^5}{2^5} = 25 = (d+1)^2. \quad (113)$$

Proof. The hidden B_1 isotypic sector of the 56-dimensional Hodge complex at $\lambda = 4$ has dimension 5, decomposing as $3_{\Omega^1} + 2_{\Omega^2} = (d-1) + (d/2) = d+1$. Its bilinear trace is $5^2 = 25 = (d+1)^2$. The Faddeev-Popov fluctuation determinant on the B_1 isotypic sector of Ω^0 satisfies the exact identity $\det(\Delta - 4I)|_{B_1 \subset \Omega^0, \text{nonzero}} = 2^5 = 32$. This equals exactly the square root of the one-loop determinant $\det(D^2|_{V_{B_1}^{\text{hidden}}}) = 4^5 = 1024$ on the 5-dimensional hidden B_1 sector. The ratio is exactly 1. Hence $C_{\text{kin}} = 25 \cdot 1 = 25$. B_1 is the unique D_4 irrep satisfying the Faddeev-Popov cancellation identity; the stress test ratios for A_1, A_2, B_2, E are 64, $-1/8$, $1/2$, and 128 respectively. $\square$

Given the disclosed physical normalization $M_P^2 = NkL^2\lambda_L^2$, the Newton constant is

$$N = \frac{25}{8\pi M_P^2}, \quad (114)$$

with the 25 derived from the Faddeev-Popov path integral and the 8π derived from the Faddeev-Popov measure on the discrete torus.

8 Level 6: Cosmology

8.1 The kL Crossover

Theorem 8.1 (kL Crossover). *The bare warp factor is*

$$kL_{\text{bare}} = \frac{(d-1)N}{d+1} = \frac{3 \cdot 64}{5} = \frac{192}{5} = 38.4 \quad (115)$$

exactly at $d = 4$, $N = 64$. The physical warp factor is $kL_{\text{phys}} = 38.442527$, matching $\ln(M_P/v_{\text{EW}})$ to 0.004%. The correction is

$$kL_{\text{phys}} - kL_{\text{bare}} = \frac{|D_4| - 1}{2} \delta_{\text{edge}} = \frac{7}{2} \delta_{\text{edge}}. \quad (116)$$

Derivation sketch. The mass-map self-consistency condition $B_{\text{mass}}(N/4)_\nu = kL - \Delta$ with offset $\Delta = 2(d-1)$ is a quadratic in kL . At $d = 4$, $N = 64$ the discriminant is exactly $121/256$ with square root $11/16$, and the physical root is $(d-1)N/(d+1) = 192/5 = 38.4$. Among $1 \leq d \leq 8$, only $d = 4$ produces a rational discriminant with $\Delta = 2(d-1)$. $\square$

The offset $\Delta = 2(d-1)$ is derived from the geometric-mean structure of the mass map: the geometric mean of the neutrino mass and the Planck mass sits $d-1$ factors of e above the geometric mean of the map top and the electroweak scale.

The geometric-mean identity is

$$m_\nu M_P = A_{\text{mass}} v_{\text{EW}} \exp(2(d-1)). \quad (117)$$

The scale λ_L introduced in the gravity sector is derived as

$$\lambda_L = \frac{(d+1)M_P}{(d-1)L^3} = \frac{5M_P}{1536}. \quad (118)$$

8.2 Dark Energy and the DME Entropy-Well Normalization

Theorem 8.2 (Dark Energy Density). *The dark energy density of HCSM is*

$$\rho_{\text{DE}} = \frac{|F|}{|B_{\text{bos}}|} m_\nu^4 = \frac{d+1}{2} m_\nu^4 = 2.4809 \times 10^{-47} \text{ GeV}^4, \quad (119)$$

matching observation to 0.76%.

Proof. The residual vacuum energy of the self-canceling torus is carried by the 14 self-paired modes. The processor T splits these into $|F| = 10$ fermionic and $|B_{\text{bos}}| = 4$ bosonic modes. The dark energy density is the ratio of fermionic to bosonic mode counts times the fourth power of the lightest fermion mass, because the neutrino is the lightest mode in the fermionic sector and therefore the dominant contributor at the observed scale. $\square$

The refined formula is

$$\rho_{\text{DE}} = \frac{d+1}{2} (1 + c\delta) m_\nu^4 = 2.5003 \times 10^{-47} \text{ GeV}^4, \quad (120)$$

matching observation to 0.010%, where $c = 5/9$ (HCSM-52, EFT Coefficient Theorem) and $\delta = (1 - \gamma)^2 / (2\pi)$ (HCSM-54).

Theorem 8.3 (DME Entropy-Well Normalization). *The DME entropy-well normalization is*

$$S_0 = v_{\text{EW}}^2 \cdot \frac{2d-1}{|F| \cdot |B_{\text{bos}}|} \cdot \left(1 + \frac{\delta_{\text{edge}}}{|F| - |B_{\text{bos}}|} \right) = v_{\text{EW}}^2 \cdot \frac{7}{40} \left(1 + \frac{\delta_{\text{edge}}}{6} \right) = 10630.722 \text{ GeV}^2, \quad (121)$$

matching the previously implied value at relative residual 2.44×10^{-6} .

The equation of state is $w = -1$ exactly at leading order, from the topological nature of the residual vacuum energy. The EFT correction is $w = -1.014054 \dots$

8.3 Vacuum Stability

Theorem 8.4 (Vacuum Stability). *The stability coefficient is*

$$\gamma = \frac{c_2}{I(2)} = 0.702839467990 \dots \quad (122)$$

with $I(2) = 1 - \frac{1}{2} \text{AGM}(1, \sqrt{2})$, $\text{AGM}(1, \sqrt{2}) = 1/(\sqrt{2}\eta(i)^2)$, and $\eta(i) = \Gamma(1/4)/(2\pi^{3/4})$. The stability theorem states $S''(0)/S_0 = -2\gamma < 0$: the equilibrium $D = 4$ is a strict local maximum of the vacuum entropy.

The bridge identity linking the stability coefficient to the Higgs quartic coupling is

$$\lambda_{\text{tree}} = \frac{\gamma^2}{d} \left(1 - \frac{\delta_{\text{edge}}}{2(d-1)} \right), \quad \gamma = \sqrt{d\lambda_{\text{tree}}} \left(1 + \frac{\delta_{\text{edge}}}{4(d-1)} \right), \quad (123)$$

holding to 8.4×10^{-8} in λ_{tree} and 1.6×10^{-6} in γ (HCSM-54, Theorem 5.2).

8.4 The Λ -Promotion Mechanism

Definition 8.5 (EFT Layer). The EFT layer L_{EFT} of the five-layer loop acts on the residual subspace κ_{14} as a scalar operator:

$$L_{\text{EFT}}|_{\kappa_{14}} = (1 + c\delta)I_{14}, \quad (124)$$

with $c = 5/9$ fixed by the Kernel-Diagonal Theorem (HCSM-14.a) and $\delta = (1 - \gamma)^2/(2\pi)$.

Theorem 8.6 (Λ -Promotion as Pure Projection). *The residual vacuum energy ρ_{res} of the 14-mode multiplet at $\lambda = 4$ is promoted to the observed cosmological constant ρ_{DE} by pure projection onto the timelike direction of $\ker D$:*

$$\rho_{\text{DE}} = \frac{d+1}{2}(1 + c\delta)m_\nu^4, \quad c = \frac{5}{9}, \quad \delta = \frac{(1 - \gamma)^2}{2\pi}. \quad (125)$$

The promotion weight is $1 + c\delta$; the projection factor onto the timelike direction of $\ker D$ is 1 (the residual is a scalar); and no gauge mediator or gravitational suppression factor is present.

Proof. Three structural facts combine. (1) The EFT layer acts as a soft projector with weight $1 + c\delta$ on the residual subspace (HCSM-56, Theorem 7). (2) The hidden $u(1)$ VEVs vanish exactly: the hidden vacuum is the D_4 -invariant state, and the generators are traceless (HCSM-56a, Theorem 4). (3) The projection of a scalar onto the 1-dimensional timelike direction of $\ker D$ has unit weight; no $1/M_P^2$ suppression is present (HCSM-56a, Theorem 7). Composing the three factors gives the promotion. $\square$

Combined with HCSM-41's result that $\epsilon \equiv 0$ exactly (no dark photon), this forces the promotion to be gravitational/geometric rather than gauge-mediated. The observed cosmological constant is $\rho_{\text{DE}} = ((d+1)/2)(1 + 5\delta/9)m_\nu^4 = 2.5003 \times 10^{-47} \text{ GeV}^4$, matching observation to 0.010%.

9 Continuum Limits and Emergent Physics

9.1 The Tensor Product Theorem

Theorem 9.1 (Tensor Product Theorem). *Let V_{14} be the 14-mode multiplet at $\lambda = 4$ on $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$, with D_4 decomposition*

$$V_{14} = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E. \quad (126)$$

Let $\ker D$ be the kernel of the Hodge-Dirac operator, with D_4 decomposition $\ker D = 2A_1 \oplus E$. Then the 56-dimensional Hodge lift $V_{56} = \Omega^0|_{\lambda=4} \oplus \Omega^1|_{\lambda=4} \oplus \Omega^2|_{\lambda=4}$ satisfies

$$V_{56} \cong V_{14} \otimes \ker D \quad (127)$$

as D_4 representations.

Proof. Compute the tensor product using the D_4 tensor product rules:

$$V_{14} \otimes \ker D = V_{14} \otimes (2A_1 \oplus E) = 2V_{14} \oplus (V_{14} \otimes E). \quad (128)$$

First, $2V_{14} = 6A_1 + 2A_2 + 4B_1 + 4B_2 + 6E$. Second, using $E \otimes E = A_1 \oplus A_2 \oplus B_1 \oplus B_2$ and $A_i \otimes E = B_j \otimes E = E$,

$$V_{14} \otimes E = 3A_1 \oplus 3A_2 \oplus 3B_1 \oplus 3B_2 \oplus 8E. \quad (129)$$

Adding gives $V_{14} \otimes \ker D = 9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E$, matching the D_4 decomposition of V_{56} . $\square$

The tensor product decomposition says: V_{14} is the internal space (the 14 modes at $\lambda = 4$ carry the Standard Model generation content); $\ker D$ is the spacetime tangent space (the 4-dimensional kernel of the Hodge-Dirac operator, with the Lorentzian metric of signature $(3, 1)$); V_{56} is the arena of the physical world.

The Bundle Construction Theorem: the physical fields of HCSM are sections of a principal bundle $P \rightarrow \ker D$ with structure group G whose Lie algebra is $\mathfrak{g}_{\text{SM}} = \mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$. The bundle is the unique D_4 -equivariant bundle over $\ker D$ with fiber V_{14} .

9.2 Maxwell-Lorentz Limit

Theorem 9.2 (Maxwell-Lorentz Limit). *The classical Maxwell-Lorentz sector of HCSM is derived from the discrete substrate. After electroweak symmetry breaking, the unbroken $\mathfrak{u}(1)_{\text{EM}}$ is identified. The photon is its gauge boson:*

$$A_\mu = \sin \theta_W W_\mu^3 + \cos \theta_W B_\mu, \quad (130)$$

with $\sin^2 \theta_W(M_Z) = 0.2312$ matching observation to 0.02%. The W and Z acquire Proca masses $m_W = 80.35 \text{ GeV}$, $m_Z = 91.63 \text{ GeV}$, with mass ratio $m_W/m_Z = \cos \theta_W = 0.8769$ matching PDG to 0.5%.

The kinematic continuum limit is derived from the Hodge star on $\ker D$. The Hodge star restricted to the 4-dimensional kernel has eigenvalues $\{+1, -1, +i, -i\}$, yielding a canonical Lorentzian metric of signature $(3, 1)$. The isotropic dispersion relation $\lambda(n) = (\pi^2/16)(n_1^2 + n_2^2) + O(n^4)$ is derived. The asymptotic expansion of the discrete exterior calculus operators is computed explicitly: $d_0/a \rightarrow \partial_\mu$ and $d_0^*d_0/a^2 \rightarrow \Delta$ with corrections of order a^2 .

The explicit $O(\epsilon^2)$ correction to the Yang-Mills action is derived:

$$S_{\text{YM}}^{(2)} = -\frac{\epsilon^2}{48} \int d^4x \text{Tr}(\partial_\lambda F_{\mu\nu} \partial^\lambda F^{\mu\nu}). \quad (131)$$

The $O(\epsilon)$ correction vanishes identically. The coefficient $1/48$ is derived from the Taylor expansion of the plaquette variable and is preserved by the tensor product structure.

9.3 Newtonian Gravity

Theorem 9.3 (Newtonian Gravity). *The Newtonian limit of HCSM gives the Newtonian potential $\phi(r) = -_NM/r$ and the Poisson equation $\nabla^2 \phi = 4\pi_N \rho_m$, with the mass density $\rho_m(x) = \sum_i m_i |\phi_i(x)|^2$ explicitly constructed from the HCSM mass map.*

The geodesic equation $d^2 x^\mu / d\tau^2 + \Gamma_{\nu\rho}^\mu \dot{x}^\nu \dot{x}^\rho = 0$ is derived from the HCSM matter action (the discrete nonlinear Schrödinger equation on the torus) via minimal coupling to the emergent metric. The equivalence principle—gravitational charge equals inertial mass—is proved as a theorem, not a postulate, using D_4 Schur orthogonality of the universal graviton-matter coupling.

The explicit $O(\epsilon^2)$, $O(\epsilon^4)$, and $O(\epsilon^6)$ corrections to the Newtonian potential are

$$\phi(r) = -\frac{NM}{r} \left[1 + \frac{1}{18} \epsilon^2 - \frac{1}{100} \epsilon^4 - \frac{331}{44100} \epsilon^6 + O(\epsilon^8) \right]. \quad (132)$$

The gauge-fixing completeness theorem (no Gribov copies on the discrete torus) and the speed-of-gravity theorem ($c_g = c_\gamma = c$ exactly) are proved.

The scale λ_L is derived from framework integers: $\lambda_L = (d+1)M_P/[(d-1)L^3] = 5M_P/1536$.

9.4 First Post-Newtonian Corrections

Theorem 9.4 (1PN Corrections). *All parametrized post-Newtonian (PPN) parameters take their general-relativistic values:*

$$\gamma = \beta = 1, \quad \alpha_1 = \alpha_2 = \alpha_3 = 0, \quad \zeta_1 = \zeta_2 = \zeta_3 = \zeta_4 = 0, \quad \xi = 0. \quad (133)$$

The Nordtvedt parameter vanishes exactly: $\eta_N = 0$.

The 1PN metric, the Einstein-Infeld-Hoffmann Lagrangian, the 1PN equations of motion, and the PPN predictions for perihelion precession, light deflection, Shapiro time delay, gravitational redshift, gravitational-wave power, and Lense-Thirring precession are all derived with $N = 25/(8\pi M_P^2)$.

The lattice corrections enter at $O(\epsilon^2)$ with $\epsilon = a/\ell_{\text{obs}}$ the lattice-to-observation scale ratio. At all currently accessible scales, $\epsilon^2 \lesssim 10^{-70}$, so the HCSM 1PN predictions are indistinguishable from general relativity.

9.5 Relativistic Energy-Momentum Limit

Theorem 9.5 (Relativistic Energy-Momentum). *The four-momentum vector $p^\mu = (E/c, \mathbf{p})$ constructed from the HCSM-native momentum operator and energy functional satisfies the mass-shell condition*

$$p^\mu p_\mu = -m^2 c^2 \quad (134)$$

exactly in the continuum limit, with m the HCSM mass-map output. The relativistic dispersion relation is

$$E^2 = p^2 c^2 + m^2 c^4. \quad (135)$$

The rest-frame limit is $E = mc^2$.

The Lorentz transformation properties of p^μ , the relativistic Doppler shift, the relativistic addition of velocities, the energy-momentum tensor $T^{\mu\nu}$, and its conservation $\partial_\mu T^{\mu\nu} = 0$ are all derived.

The DNLS coupling closes to $g = q = 33/64$ from the Hartree trace $\text{Tr}(K) = 11/32$ and the mass-map coefficient $(d-1)/2 = 3/2$.

9.6 Dirac Equation from the Hodge-Dirac Operator

Theorem 9.6 (Dirac Equation). *The Hodge complex at each site, $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$, has real dimension $1 + 2 + 1 = 4$, matching the dimension of a four-component spinor. The Fourier symbol $M(p)$ of the Hodge-Dirac operator satisfies $M(p)^2 = |p|^2 I_4$. The two substrate momentum operators A_1, A_2 generate a Clifford algebra $\text{Cl}(3, 1)$ with the additional generators $A_3 = A_1 A_2$ and $A_0 = i A_3$. The chirality operator $\gamma_5 = -i A_1 A_2$ is unitarily equivalent to the Hodge form-parity $(-1)^F$. The continuum limit is*

$$\frac{D}{a} \rightarrow i\gamma^\mu \partial_\mu + O(\epsilon^2). \quad (136)$$

The mass term is supplied by the HCSM mass map. The mode-dependent Dirac equation on the 56-dimensional sector is

$$(D_{56} + m_i)\psi_i = 0, \quad i = 1, \dots, 14, \quad (137)$$

where m_i is the mass-map output for mode i . In the continuum limit this becomes $(i\gamma^\mu \partial_\mu + m_i)\psi_i = 0$.

The Dirac bilinears (scalar, vector, tensor, axial vector, pseudoscalar) are constructed from the Hodge fiber. The spinor representation of $SO(3, 1)$ is derived from the Hodge fiber. Particle-antiparticle structure and Majorana reduction are constructed from the substrate.

9.7 Direct Quantization of the Hodge Complex

Theorem 9.7 (Direct Quantization). *The Hodge complex $\Omega^\bullet(\Lambda)$ with the Hodge inner product is a real Hilbert space $\mathcal{H}_{256}$ of dimension 256. The Hodge-Dirac operator $D = d + d^*$ is self-adjoint on $\mathcal{H}_{256}$. The full spectrum of D is computed explicitly. The $\mathbb{Z}_2$ grading $(-1)^F$ determines the statistics: even-degree forms are bosonic, odd-degree forms are fermionic. The Fock space is*

$$\mathcal{F} = \text{Sym}(\mathcal{H}_+) \otimes \Lambda(\mathcal{H}_-), \quad (138)$$

with $\dim \Lambda(\mathcal{H}_-) = 2^{128}$. The propagator is $G(p) = (M(p) - i\epsilon)^{-1}$, reducing to the Dirac propagator in the continuum limit. The path integral measure is well-defined and unique on the finite lattice. Ward identities follow from the $\mathbb{Z}_2$ symmetry.

No intermediate classical field theory is used, and no second-quantization postulate is added. The quantum field theory is the Hodge complex itself.

10 Falsifiable Predictions

1. **No light dark photon.** The kinetic mixing parameter satisfies $\epsilon = 0$ exactly. Falsifiable by SHiP, FASER2, MATHUSLA, CODEX-b, LDMX, NA64, Belle II, and HL-LHC at $\epsilon > 10^{-12}$.
2. **Strong CP.** $\theta_{\text{QCD}} = 0$ exactly. Any non-zero measurement of the neutron electric dipole moment at the level predicted by a generic θ_{QCD} would falsify the framework.
3. **ρ parameter.** Tree-level $\rho = m_W^2 / (m_Z^2 \cos^2 \theta_W) = 1$ exactly. Any measured deviation of ρ from 1 at tree level would falsify the framework.
4. **Fermion content.** One Standard Model generation contains exactly 14 particles. A fourth generation would falsify the framework.
5. **Kernel structure.** The winding kernel has dimension 10 on Λ_8 and the kernel-diagonal values are $\{1/2, 3/4\}$. Any modification of the spectrum of H_{wind} that changes these values would falsify the framework.
6. **Dark energy.** The dark energy density is $\rho_{\text{DE}} = ((d+1)/2)(1+c\delta)m_\nu^4$, matching observation to 0.010%. Any future measurement of ρ_{DE} that deviates from this formula at more than 0.1% would falsify the framework.
7. **Charge rigidity.** The charge assignment on the 14-mode multiplet satisfies exactly two structurally consistent configurations, both with total charge $-1e$. A measurement of the vacuum's total electromagnetic charge that differs from $-1e$ per one Standard Model generation would falsify the rigidity theorem.
8. **Identification.** The Midpoint Sorting Theorem assigns the 14 modes uniquely. A Standard Model particle with an $N/4$ label outside the set $\{60, 24, 21, 20, 14, 14, 12, 9.5, 9, 7.5, 2, 2, 1, 0.5\}$ would falsify the framework.
9. **A_1 uniqueness.** The physical color algebra is the $\mathfrak{su}(3)$ subalgebra on the A_1 isotypic block, forced by the dimension-and-invariance theorem. A different embedding of $\mathfrak{su}(3)_c$ in the D_4 commutant would falsify Theorem 5.4 of HCSM-06.

10. **Hypercharge values.** The hypercharge values are uniquely determined by the content structure and anomaly cancellation. A different set of hypercharge values satisfying the same content structure and anomaly conditions would falsify Theorem 8.1 of HCSM-07.
11. **Mirror cancellation.** The Laplacian spectrum is exactly mirror-symmetric under $\lambda \leftrightarrow 8 - \lambda$ and the residual ratio is exactly $7/32$. Any deviation from these exact structural identities would falsify the mirror-cancellation theorem.
12. **TF grading and Klein orbit.** The TF operator grades the 14-mode multiplet by $|\Delta n|$, and the four modes $\{u, s, \text{dark}, \tau\}$ carry a faithful Klein four-group action. Any modification of these exact structural identities would falsify HCSM-10.
13. **5+9 split.** The 14-mode multiplet splits into exactly 5 Substrate Eigenmodes and 9 Mesh Resonances. Any modification of the derived base formula or shell values would falsify HCSM-11.
14. **Layer independence.** The 14-mode multiplet carries four independent structural layers; layers 1, 2, 3 are HCSM-native and mutually compatible, while layer 4 requires external calibration. Any HCSM-native derivation of the processing layer shape parameters r_a would falsify the layer-independence theorem.
15. **Entropic flatness.** The natural state on the 56-dimensional Hodge sector is the maximally mixed state $\rho_{\text{in}} = I_{56}/56$, with entropy reduction $\Delta S = \log(7/5)$. Any deviation from these exact structural identities would falsify the Hodge-HCSM interface.
16. **Loop and DNLS commutants.** The loop commutant is 3-dimensional abelian; the DNLS commutant is 10-dimensional, $\mathfrak{su}(2)_L + \mathfrak{u}(1)^7$. Any deviation from these dimensions would falsify HCSM-15.
17. **Doubled-momentum pairing.** The B_1 -compressed kernel vanishes exactly: $K_{B_1} = 0$. Any nonzero value would falsify Theorem 5.26.
18. **Chirality-breaking localization.** The commutator $[T, P_{rs}]$ is supported exactly on $\{\text{dark}, \tau\}$. Any additional support would falsify HCSM-13.
19. **Graviton masslessness.** The graviton is exactly massless; no fifth force, no scalar graviton, no Kaluza-Klein tower. Any detection of a discrete graviton excitation spectrum at accessible energies would falsify the framework.
20. **Speed of gravity.** $c_g = c_\gamma = c$ exactly. Any confirmed detection of $c_g \neq c_\gamma$ would falsify HCSM-62.
21. **PPN parameters.** $\gamma = \beta = 1$ exactly; $\eta_N = 0$ exactly. Any confirmed deviation at the 10^{-5} level would falsify the framework.
22. **Mass-shell condition.** $p^\mu p_\mu = -m^2 c^2$ exactly. Any confirmed deviation would falsify HCSM-64.
23. **Relativistic dispersion.** $E^2 = p^2 c^2 + m^2 c^4$ exactly. Any confirmed deviation would falsify HCSM-64.
24. **Cabibbo angle.** $\sin \theta_{12} = \pi/14 = 0.2243995$ exactly; a 10^{-4} deviation falsifies HCSM-34.

25. **$|V_{td}|$ prediction.** $|V_{td}| = 0.0061 \pm 0.0001$, testable by Belle II and LHCb in 2026–2028.
26. **PMNS angles.** $\theta_{12} = 34.2^\circ$, $\theta_{23} = 47.9^\circ$, $\theta_{13} = 8.5^\circ$; 0.5° deviations falsify HCSM-36.
27. **Generation phase.** $\phi_{\text{gen}} = 0.6\pi$ exactly. Any confirmed deviation falsifies HCSM-37.
28. **Normal ordering.** The neutrino mass ordering is normal; inverted ordering would falsify HCSM-36.
29. **Higgs mass.** $m_H = 125.138$ GeV given v_{EW} ; a refinement of v_{EW} tests HCSM-40.
30. **Top mass.** $m_t = 172.688398$ GeV. An improved m_t measurement at the 0.001% level tests HCSM-40.
31. **Bridge identity.** A refined independent measurement of γ and λ_{tree} tests the 8.4×10^{-8} identity.
32. **kL crossover.** $kL_{\text{bare}} = 38.4$ exactly. The rational discriminant $121/256$ is unique to $d = 4$ among small dimensions.
33. **Equation of state.** $w = -1$ exactly at leading order; EFT correction $w = -1.014054\dots$ testable at $> 3\sigma$ by DESI DR3 + Euclid.
34. **DME entropy-well normalization.** $S_0 = 10630.722$ GeV² with residual 2.44×10^{-6} .
35. **Λ -promotion.** The refined prediction is $\rho_{\text{DE}} = 2.5003 \times 10^{-47}$ GeV⁴. A measured value outside $[2.4995, 2.5015] \times 10^{-47}$ GeV⁴ falsifies the projection mechanism.
36. **Λ -promotion weight.** The timelike projection weight is exactly 1; a deviation to $1/4$ or any other value would falsify HCSM-56a.
37. **Hidden VEVs vanish.** A confirmed nonzero hidden $\mathfrak{u}(1)$ VEV would falsify HCSM-56a.
38. **Newton constant.** $N = 25/(8\pi M_P^2)$ with $M_P^2 = NkL^2\lambda_L^2$. Matches observation to 0.14%.
39. **λ_L derivation.** $\lambda_L = 5M_P/1536$. Any deviation from this value would falsify HCSM-62.
40. **No KK tower.** The absence of a Kaluza-Klein tower is a sharp prediction: any detection of a discrete graviton excitation spectrum at accessible energies would falsify HCSM-46.
41. **B_2 silence.** B_2 does not contribute to the two-loop running of the Weinberg angle. If future computation shows a nonzero B_2 contribution, Proposition 7.1 of HCSM-48 is falsified.

11 Theorem Inventory

The framework’s theorems are catalogued by their home paper. New theorems from the present revision cycle are marked with *.

Theorem	Location
Laplacian uniqueness	HCSM-01 Thm 2.1
$d \cdot 2^d = 64$, $L = 8$	HCSM-01 Cor 4.1
Two derivations of $L = 8$	HCSM-01 S4, HCSM-14.b

Theorem	Location
$2(L - 1) = 14$	HCSM-01 Thm 4.4
Unified substrate selection	HCSM-00 S2.6
Single lattice primitive*	HCSM-00 S2.7
Mirror symmetry*	HCSM-00 Thm 2.16
$50 + 14$ decomposition*	HCSM-00 Cor 2.19
Residual ratio $7/32$ *	HCSM-00 Thm 2.22
TF grading by $ \Delta n $ *	HCSM-00 Thm 2.28
Klein four-orbit*	HCSM-00 Thm 2.32
Hodge-Dirac square	HCSM-02 Thm 3.2
Lorentzian signature	HCSM-02 Thm 3.9
Kernel dimension 4	HCSM-02 Thm 6.1
Spectral gap $2\sin(\pi/L)$	HCSM-02 Thm 6.4
D_4 decomposition of V_{56}	HCSM-02 Thm 7.1
Self-paired eigenvalue	HCSM-04 Thm 3.2
Explicit mode list	HCSM-04 Thm 4.1
D_4 orbit decomposition	HCSM-04 Thm 5.1
Support classes $\{64, 48, 32\}$	HCSM-04 Prop 6.2
IPRs $\{2/128, 3/128, 4/128\}$	HCSM-04 Prop 7.2
Affine entropy-IPR relation	HCSM-04 Thm 8.2
Entropy classes $\{6.0, 5.5, 5.0\}$	HCSM-04 Cor 8.3
D_4 character χ_{14}	HCSM-05 Thm 4.1
Irreducible decomposition	HCSM-05 Thm 6.1
Joint (D, T) class partition*	HCSM-05 Thm 7.7
Isotypic projector degeneracy (loop)*	HCSM-15 Thm 4.1
No root generators (loop)*	HCSM-15 Thm 4.3
A_1 uniqueness for color	HCSM-06 Thm 5.4
B_1 / B_2 outer automorphism	HCSM-06 Thm 5.10
Sector-coarse 15-dim space	HCSM-06 Thm 6.3
A_1 hypercharge coefficients	HCSM-07 Thm 5.1
B_1 hypercharge coefficients	HCSM-07 Thm 6.1
Hypercharge quantization	HCSM-07 Thm 7.1
Hypercharge values from anomaly	HCSM-07 Thm 8.1
Content structure from HCSM data	HCSM-07 Thm 8.2
Orientation-flip $\mathbb{Z}_2$	HCSM-09 Thm 5.1
Sector-operator transformation	HCSM-09 Thm 6.1
Mirror cancellation	HCSM-10 Thm 3.4
Residual ratio $7/32$	HCSM-10 Thm 7.1
Convention independence	HCSM-10 Thm 8.1
Derived base formula*	HCSM-11 Thm 3.1
$5 + 9$ split*	HCSM-11 Thm 4.2
Shell formula*	HCSM-11 Thm 5.1
W/Z base formula*	HCSM-11 Thm 6.1
Mirror residual as Hartree eigenvalue*	HCSM-11 Thm 7.2
T_{half} commutes with Hartree*	HCSM-11 Thm 7.3
Coefficient 26 *	HCSM-11 Thm 8.1
Layer independence*	HCSM-11 Thm 9.1

Theorem	Location
Entropic flatness	HCSM-12 Thm 3.1
Processor channel output	HCSM-12 Thm 4.2
Entropy reduction $\log(7/5)$	HCSM-12 Cor 5.1
Fermion/Boson-Class Correspondence	HCSM-13 Thm 3.1
Chirality-Breaking Localization*	HCSM-13 Thm 4.1
Observer pair $(W, t)^*$	HCSM-13 S5.3
Kernel-Diagonal Theorem	HCSM-14 Thm 7.3
Kernel- T decomposition*	HCSM-14 Thm 8.1
Family-block eigenvalues*	HCSM-14 Thm 10.1
Four kernel constants*	HCSM-14 Thm 11.1
Rank formula*	HCSM-14 Thm 9.3
Loop commutant*	HCSM-15 Thm 3.3
DNLS commutant*	HCSM-15 Thm 5.4
Doubled-momentum pairing*	HCSM-15 Thm 6.2
Exact $\mathfrak{su}(2)_L$ preservation*	HCSM-15 Cor 6.3
$\mathfrak{su}(3)_c$ breaking*	HCSM-15 Thm 7.1
Explicit B_1 basis vectors*	HCSM-15 App A
Loop commutant universality	HCSM-16 Thm 10
Hidden-sector neutrality	HCSM-16 Thm 11
DNLS commutant universality	HCSM-16 Thm 15
Extension map	HCSM-16 Lemma 3.2
Cartan rotation angles	HCSM-17 Thm 6.2
DNLS on torus	HCSM-18 Thm 4
Class-homogeneity of K	HCSM-18 Thm 14
IPR diagonal of K	HCSM-18 Thm 11
Doubled-momentum pairing	HCSM-18 Thm 17
$(d-1)/2$ coefficient	HCSM-19 Thm 6.1
Hartree orbit trace identity	HCSM-19 Thm 7.1
Motion Theorem	HCSM-20 Thm 13.1
H_{wind} characteristic polynomial	HCSM-20 Thm 3.4
Standing wave amplitudes	HCSM-20 Thm 3.8
Closed form loop eigenvalues	HCSM-20 Thm 9.2
Exact ratio $ M_A / M_B = 3$	HCSM-20 Thm 9.4
Wick rotation rate	HCSM-20 Thm 10.1
Lattice corrections	HCSM-20 Thm 11.1
Time unit $T_{\text{unit}} = 2\pi/m_s$	HCSM-20 Thm 12.1
Zero state uniqueness	HCSM-22 Thm 4
Zero state vector	HCSM-22 Thm 7
rs -oddness of zero state	HCSM-22 Thm 8
Exact ratios of zero state	HCSM-22 Thm 12
Zero-state selection principle	HCSM-22a Thm 3.1
Landauer route to selection	HCSM-22a Thm 4.1
Entropy-well route to selection	HCSM-22a Thm 5.1
Route equivalence	HCSM-22a Thm 6.1
Midpoint Sorting Theorem	HCSM-23 Thm 5.2
Within-pair ordering rule	HCSM-23 Thm 4.3

Theorem	Location
nu/e resolution	HCSM-23 Cor 4.4
$N/4$ labels from framework constants*	HCSM-23 Rev. 8 Thm 4.1
(S, D) table as corollary*	HCSM-23 Rev. 8 Cor 4.4
Charge rigidity class	HCSM-24 Thm 6.3
Charge assignment solutions	HCSM-24 Thm 6.4
Total charge rigidity	HCSM-24 Thm 6.5
Orbit-Class Reconstruction	HCSM-28 Thm 3
Distinguished-Pair Graph Decomposition	HCSM-28 Thm 4
Charge-Entropy-Observer Triangle	HCSM-29 Thm 1.1
Triangle trace identity	HCSM-29 Thm 6.2
Base-2 pattern	HCSM-30 Thm 3.2
Coefficient 26	HCSM-30 Thm 2.2
Crossover at kL	HCSM-30 Thm 6.1
Generation count $ A = d - 1 = 3$	HCSM-31 Thm 17.1
Flavor weights	HCSM-31 Thm 16.2
Homogeneity constraint	HCSM-31 Thm 14.3
Shell assignment rule	HCSM-32 Thm 4.2
Support values $\{64, 48, 32\}$	HCSM-32 Thm 3.2
Mass map	HCSM-33 Thm 3.1
Geometric-mean identity	HCSM-33 Thm 5.1
Neutrino mass $m_\nu = 1.775$ meV	HCSM-33 Thm 4.2
Cabibbo angle $\sin \theta_{12} = \pi/14$	HCSM-34 Thm 1
Wolfenstein parameters	HCSM-34 Thm 3
Rank 6 dissolution	HCSM-34 Thm 7
Fermion-excess correction	HCSM-34 Thm 9
$\mathbb{Z}_3$ cyclic partition	HCSM-35 Thm 5.1
Daughter relations	HCSM-35 Thm 6.2
Residual-bath coupling 50/51	HCSM-35 Thm 7.1
Phase relation for V_{td}	HCSM-35 Thm 8.1
Mirror identity	HCSM-35 Thm 9.1
Anti-daughter relations	HCSM-35 S10
PMNS from spin structures	HCSM-36 Thm 4.1
Neutrino $N/4$ spectrum (60, 60, 58)	HCSM-36 Thm 4.1
Normal ordering	HCSM-36 Thm 4.3
Mixing angles	HCSM-36 Thm 6.1
CKM/PMNS asymmetry	HCSM-36 Thm 7.1
Generation phase $\phi_{\text{gen}} = 0.6\pi$	HCSM-37 Thm 3.1
Unification with CKM amplitude	HCSM-37 Thm 4.1
Higgs doublet	HCSM-40 Thm 3.2
Weak direction from zero state	HCSM-40 Thm 4.1
DNLS $\mathfrak{su}(2)_L$ preservation	HCSM-40 Thm 5.1
Two-loop anomaly closure	HCSM-40 Thm 6.3
$\mathcal{O}_2$ self-mirror orbit	HCSM-40 Thm 7.1
Top-Yukawa coefficient 4/3	HCSM-40 Thm 7.2
Quartic coupling λ_{tree}	HCSM-40 Thm 8.1
Bridge identity	HCSM-40 Thm 9.1

Theorem	Location
Single mass anchor	HCSM-40 Thm 10.1
Absence of light dark photon	HCSM-41 Thm 3.2
No tree-level kinetic mixing	HCSM-41 Thm 4.1
No one-loop kinetic mixing	HCSM-41 Thm 5.1
Exact $\epsilon = 0$	HCSM-41 Cor 6.1
Falsification threshold	HCSM-41 Thm 7.1
Hodge form-parity γ_5	HCSM-42 Thm 3.2
56-dim splitting $\mathcal{B} \oplus \mathcal{F}$	HCSM-42 Thm 4.1
Spin-structure parity	HCSM-42 Thm 5.2
Compatibility $[F, \gamma_5] = 0$	HCSM-42 Thm 6.2
Compatibility $[F, \chi_{\text{spin}}] = 0$	HCSM-42 Cor 6.5
$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ structure	HCSM-42 Thm 7.1
Spin-statistics consistency	HCSM-42 Thm 8.1
Orientation $\mathbb{Z}_2$ action	HCSM-43 Thm 3.1
Exact anticommutation $\{F, H_{\text{wind}}\} = 0$	HCSM-43 Thm 4.1
Strong CP $\theta_{\text{QCD}} = 0$	HCSM-43 Thm 5.3
$\theta_{\text{eff}} = 0$	HCSM-43 Cor 6.2
$\mathfrak{u}(1)_Y$ origin	HCSM-44 Thm 3.2
Proca Lagrangian	HCSM-44 Thm 4.1
Mass ratio $m_W/m_Z = \cos \theta_W$	HCSM-44 Thm 5.1
Transversality	HCSM-44 Thm 6.1
$\rho = 1$ at tree level	HCSM-44 Cor 7.1
Symmetric-square decomposition	HCSM-45 Thm 3.4
Graviton sector $54B_1 \oplus 54B_2 = 108$	HCSM-45 Cor 4.2
Polarization space $V = B_1 \oplus B_2$	HCSM-45 Thm 5.1
Self-paired bilinear $h = \omega \otimes \omega$	HCSM-45 Thm 6.2
Graviton exact masslessness	HCSM-46 Thm 4
No finite- N correction	HCSM-46 Prop 6
Retraction of WIN-era estimates	HCSM-46 S5
Pure-polynomial counts $(0, 2, 0, 3)$	HCSM-47 Thm 3.2
Derivative-included counts $(5, 12, 11)$	HCSM-47 Thm 4.2
Odd-derivative selection rule	HCSM-47 Thm 5.1
$D_4 \subset O(3, 1)$ embedding	HCSM-47 Thm 6.2
Graviton as composite bilinear	HCSM-47 Prop 7.1
Faddeev-Popov field-independence	HCSM-47 Prop 7.2
Hidden B_1 dimension $3 + 2 = 5$	HCSM-48 Thm 3.2
Bilinear trace $5^2 = 25$	HCSM-48 Thm 3.2
FP cancellation identity $2^5 = 32$	HCSM-48 Thm 4.1
Uniqueness of B_1	HCSM-48 Thm 5.1
B_2 spectator status	HCSM-48 Prop 7.1
Kinetic coefficient $C_{\text{kin}} = 25$	HCSM-48 Thm 6.1
kL crossover	HCSM-51 Thm 7
Rational discriminant 121/256	HCSM-51 Thm 6
Offset $\Delta = 2(d - 1)$	HCSM-51 Thm 4
Uniqueness of $d = 4$	HCSM-51 Thm 9
Geometric-mean form	HCSM-51 Thm 11

Theorem	Location
EFT coefficient $c = 5/9$	HCSM-52 Thm 3.1
Dark energy density	HCSM-53 Thm 5
DME entropy-well normalization S_0	HCSM-53 Thm 7
$w = -1$ exactly	HCSM-53 Thm 10
EFT-corrected w	HCSM-53 Thm 11
AGM identity	HCSM-54 Thm 3.1
Closed form of c_2	HCSM-54 Thm 3.2
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Substrate-level AGM	HCSM-54 Thm 3.4
Crossover-zero moments 4, 12, 28	HCSM-54 Thm 4.1
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Λ -promotion as pure projection	HCSM-56a Thm 1
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Newtonian gravity	HCSM-62 Thm 4.2
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Rest-frame $E = mc^2$	HCSM-64 Thm 6.3
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DNLS coupling $g = q = 33/64$	HCSM-64 S11.5
Dirac equation from Hodge-Dirac	HCSM-65 Thm 5.2

Theorem	Location
Clifford algebra $\text{Cl}(3, 1)$	HCSM-65 Thm 3.1
Chirality operator	HCSM-65 Thm 4.2
Unitary equivalence of γ_5 and $(-1)^F$	HCSM-65 Thm 4.7
Continuum limit $D/a \rightarrow i\gamma^\mu \partial_\mu$	HCSM-65 Thm 5.2
Dirac bilinears	HCSM-65 S7
Spinor representation of $SO(3, 1)$	HCSM-65 Thm 9.1
Direct quantization	HCSM-66 Thm 2.3
Self-adjointness of D	HCSM-66 Thm 3.1
Spectrum of D	HCSM-66 Thm 3.3
Statistics from grading	HCSM-66 Thm 5.4
Fock space dimension 2^{128}	HCSM-66 Thm 7.2
Propagator	HCSM-66 Thm 9.2
Path integral measure	HCSM-66 Thm 10.1
Ward identities	HCSM-66 Thm 11.1

All rank blockers of the framework’s derivation program are closed. As of the present revision, the physical identification of the A_1 and B_1 blocks is derived (color) or a labeling convention (weak isospin); the hypercharge values and content structure are derived; the mirror cancellation, 5+9 split, Hodge-HCSM interface, orientation-flip $\mathbb{Z}_2$, processor commutant structure, doubled-momentum pairing, continuum limits (Maxwell-Lorentz, Newtonian, 1PN, special relativity, Dirac, quantization), graviton masslessness, Faddeev-Popov G_N normalization, dark energy, vacuum stability, kL crossover, and Λ -promotion are integrated. The framework has zero empirical anchors in the particle, gauge, hypercharge, mass, and dark energy sectors.

12 Reproducibility

All numerical claims in the HCSM paper stack are verified at machine precision in IEEE-754 double precision. The verification scripts use only standard libraries (NumPy, SciPy, SymPy, mpmath) and construct the objects they need from HCSM-native definitions. Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the stack.

The complete HCSM paper stack, verification scripts, and source are available at

<https://github.com/007STAN/HODGE-COMPLEX-STANDARD-MODEL-HCSM>

13 Conclusion

This paper has established the foundation of the Hodge Complex Standard Model. The substrate is the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$, selected by the unified substrate selection theorem (Section 2.6). The substrate carries the Hodge complex $\Omega^0 + \Omega^1 + \Omega^2$ with $D^2 = \Delta$ exactly on each form degree. The 14 modes at the self-paired eigenvalue $\lambda = 4$ are identified with one Standard Model generation. The processor T and its five-layer composition $M_{5\text{layer}}$ are the machine’s operation. The output structures (charge, mass, flavor, gravity, cosmology) are the residue after mirror cancellation.

The framework’s most consequential structural results are now as follows.

First, the Kernel-Diagonal Theorem unifies the framework’s three leading coefficients under a single projector: the EFT coefficient $c = 5/9$, the mass-map coefficient $(d - 1)/2 = 3/2$, and the kernel dimension $10 = (d - 1)^2 + 1$ are three traces of the diagonal of the projector onto $\ker(H_{\text{wind}})$.

Second, the Tensor Product Theorem $V_{56} \cong V_{14} \otimes \ker D$ closes the continuum limit. The physical world is the tensor product of the internal Standard Model content and the emergent spacetime.

Third, the framework derives the classical Maxwell-Lorentz sector, Newtonian gravity, first post-Newtonian corrections, the relativistic energy-momentum relation, the Dirac equation, and the direct quantization of the Hodge complex. The emergent Lorentzian metric of signature $(3, 1)$ is derived from the Hodge star on $\ker D$.

Fourth, the Λ -promotion mechanism is closed: the residual vacuum energy is promoted by pure projection onto the timelike direction of $\ker D$, and the hidden $\mathfrak{u}(1)$ VEVs vanish.

Fifth, the framework's coverage of the Standard Model gauge content is complete: absence of a light dark photon ($\epsilon \equiv 0$), spin-chirality $\mathbb{Z}_2$ grading ($\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$), strong-CP solution ($\theta_{\text{eff}} = 0$), Proca sector ($\rho = 1$ at tree level), and topological protection of ϕ_{gen} .

Sixth, the charge sector is a rigidity-class result: five structural constraints reduce the charge assignment from 5^{14} configurations to exactly two, both with total charge $-1e$, and the residual binary choice is resolved by the Midpoint Sorting Theorem.

Seventh, the identification thesis is closed: the 14 modes are uniquely sorted into the 14 Standard Model particles by native HCSM data, with zero empirical anchors remaining in the particle sector.

Eighth, the mirror-cancellation theorem is integrated into the foundations: the mirror symmetry $\lambda \leftrightarrow 8 - \lambda$ pairs 50 of the 64 scalar modes into cancelling partners, only the 14 self-paired modes at $\lambda = 4$ survive, and the residual ratio is exactly $7/32$.

Ninth, the $5+9$ split of the 14-mode multiplet and the Hodge-HCSM interface are integrated: 5 Substrate Eigenmodes and 9 Mesh Resonances; the natural state on the 56-dimensional Hodge sector is the maximally mixed state; the processor channel is a completely positive trace-preserving map; and the entropy reduction is exactly $\log(7/5)$.

Tenth, the processor commutant structure is integrated: the loop commutant is a 3-dimensional real, Hermitian, abelian subalgebra of $\mathfrak{g}_{\text{SM}}$; the DNLS commutant is 10-dimensional, isomorphic to $\mathfrak{su}(2)_L + \mathfrak{u}(1)^7$.

Eleventh, the doubled-momentum pairing mechanism is integrated: the B_1 -compressed Hartree kernel vanishes identically, $K_{B_1} = 0$, by the explicit B_1 basis vectors constructed in Appendix A.

Twelfth, the chirality-breaking localization is integrated: the processor commutator $[T, P_{rs}]$ is supported exactly on the two-element set $\{\text{dark}, \tau\}$.

Thirteenth, the $N/4$ labels are derived from framework constants: each of the fourteen labels is an elementary function of the framework integers $\{d, N, L, H\} = \{4, 64, 8, 16\}$.

Fourteenth, the graviton is exactly massless and its self-coupling structure is derived from D_4 ; the Newton constant is $N = 25/(8\pi M_P^2)$ from first principles (given the disclosed physical normalization $M_P^2 = NkL^2\lambda_L^2$).

Fifteenth, the kL crossover is derived as the exact rational number $kL_{\text{bare}} = 192/5 = 38.4$, with the rational discriminant $121/256$ unique to $d = 4$ among small dimensions.

Sixteenth, the dark energy density and the DME entropy-well normalization are derived, and the equation of state is $w = -1$ exactly at leading order.

Seventeenth, the framework now has a single dimensionful anchor (M_P) and derives every other quantity from the substrate at $d = 4$ and $L = 8$. The scale $\lambda_L = 5M_P/1536$ is derived from framework integers.

The framework's claim is that the Standard Model is what the substrate looks like when you observe it.

Physics is what the machine cannot cancel. The Hodge complex is the machine. The

$-1e$ is what the machine must not cancel. The kernel is what the machine cannot cancel.

A Explicit B_1 Basis Vectors and the Doubled-Momentum Pairing Theorem

This appendix constructs the explicit orthonormal basis vectors of the B_1 isotypic sector of the 14-mode multiplet and verifies the doubled-momentum pairing property used in Theorem 5.26.

A.1 The B_1 Isotypic Projector

The B_1 isotypic projector on the 14-mode multiplet is

$$P_{B_1} = \frac{1}{8} [R(e) + R(r^2) - R(r) - R(r^3) + R(s) + R(r^2s) - R(rs) - R(r^3s)], \quad (139)$$

where the D_4 action on mode coordinates is given by the standard permutation representation. The B_1 character is $\chi_{B_1} = (1, 1, -1, 1, -1)$ on the conjugacy classes (e, r^2, r, s, rs) .

A.2 Explicit Construction of the Basis Vectors

Proposition A.1 (First B_1 Basis Vector). *Applying P_{B_1} to the neutrino mode $\nu = (0, 4)$ gives, after normalization,*

$$v_1^{(B_1)} = \frac{|\nu\rangle - |e\rangle}{\sqrt{2}} = \frac{|(0, 4)\rangle - |(4, 0)\rangle}{\sqrt{2}}. \quad (140)$$

Proposition A.2 (Second B_1 Basis Vector). *Applying P_{B_1} to the up-quark mode $u = (1, 3)$ gives, after normalization,*

$$v_2^{(B_1)} = \frac{\sqrt{2}}{4} (|u\rangle + |\tau\rangle - |b\rangle - |\mu\rangle + |d\rangle + |c\rangle - |s\rangle - |\text{dark}\rangle). \quad (141)$$

In coordinate form:

$$v_2^{(B_1)} = \frac{\sqrt{2}}{4} (|(1, 3)\rangle + |(7, 5)\rangle - |(5, 1)\rangle - |(3, 7)\rangle + |(1, 5)\rangle + |(7, 3)\rangle - |(3, 1)\rangle - |(5, 7)\rangle). \quad (142)$$

Proposition A.3 (Orthonormality). *The two vectors $v_1^{(B_1)}$ and $v_2^{(B_1)}$ are orthonormal:*

$$v_1 \cdot v_1 = 1, \quad v_2 \cdot v_2 = 1, \quad v_1 \cdot v_2 = 0. \quad (143)$$

A.3 The Matrix U_{B_1}

In the standard mode order $(\nu, e, u, d, s, \mu, \text{dark}, c, \tau, b, W, Z, H, t)$:

$$U_{B_1} = \begin{pmatrix} 1/\sqrt{2} & 0 \\ -1/\sqrt{2} & 0 \\ 0 & \sqrt{2}/4 \\ 0 & \sqrt{2}/4 \\ 0 & -\sqrt{2}/4 \\ 0 & -\sqrt{2}/4 \\ 0 & -\sqrt{2}/4 \\ 0 & \sqrt{2}/4 \\ 0 & \sqrt{2}/4 \\ 0 & -\sqrt{2}/4 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}. \quad (144)$$

This matrix satisfies $U_{B_1}^T U_{B_1} = I_2$.

A.4 The Doubled-Momentum Pairing Structure

Proposition A.4 (Pairing Structure of v_1). *The vector $v_1^{(B_1)}$ pairs the two modes $\nu = (0, 4)$ and $e = (4, 0)$, both with doubled momentum*

$$(2 \cdot 0, 2 \cdot 4) = (2 \cdot 4, 2 \cdot 0) = (0, 0) \pmod{8}, \quad (145)$$

and with opposite signs $+1/\sqrt{2}$ and $-1/\sqrt{2}$.

Proposition A.5 (Pairing Structure of v_2). *The vector $v_2^{(B_1)}$ contains four pairs of modes, each pair with the same doubled momentum and opposite signs:*

Pair	Mode 1	Doubled Momentum	Mode 2
1	$u = (1, 3)$	$(2, 6)$	dark = $(5, 7)$
2	$\tau = (7, 5)$	$(6, 2)$	$s = (3, 1)$
3	$d = (1, 5)$	$(2, 2)$	$b = (5, 1)$
4	$c = (7, 3)$	$(6, 6)$	$\mu = (3, 7)$

A.5 Proof that $K_{B_1} = 0$

Theorem A.6 (B_1 -Compressed Kernel Vanishes). *The B_1 -compressed Hartree kernel vanishes identically:*

$$K_{B_1} = U_{B_1}^T K U_{B_1} = 0 \in \mathbb{R}^{2 \times 2}. \quad (146)$$

Proof. The compression K_{B_1} has entries

$$(K_{B_1})_{ij} = (v_i^{(B_1)})^T K v_j^{(B_1)} = \sum_x \rho_i(x) \rho_j(x), \quad (147)$$

where the density profiles are $\rho_i(x) = \sum_a v_i^{(a)} |\phi_a(x)|^2$, $i = 1, 2$. For $i = 1$: using the pairing structure of v_1 ,

$$\rho_1(x) = \frac{1}{\sqrt{2}} (|\phi_\nu(x)|^2 - |\phi_e(x)|^2). \quad (148)$$

By the density-equality lemma, $|\phi_\nu(x)|^2 = |\phi_e(x)|^2 = 1/64$ for all $x \in \Lambda$. Therefore $\rho_1(x) = 0$ for all $x \in \Lambda$. For $i = 2$: using the pairing structure of v_2 ,

$$\rho_2(x) = \frac{\sqrt{2}}{4} [|\phi_u(x)|^2 - |\phi_{\text{dark}}(x)|^2 + |\phi_\tau(x)|^2 - |\phi_s(x)|^2 + |\phi_d(x)|^2 - |\phi_b(x)|^2 + |\phi_c(x)|^2 - |\phi_\mu(x)|^2]. \quad (149)$$

By the density-equality lemma, each of the four pairs satisfies $|\phi_{m'}(x)|^2 = |\phi_m(x)|^2$. Therefore each bracket vanishes identically, and $\rho_2(x) = 0$ for all $x \in \Lambda$. Since $\rho_1 = \rho_2 = 0$ identically, every entry of K_{B_1} vanishes: $K_{B_1} = 0$. $\square$

A.6 Corollary: Exact $\mathfrak{su}(2)_L$ Preservation

Corollary A.7 ($\mathfrak{su}(2)_L$ Preservation). *The three generators of $\mathfrak{su}(2)_L$ commute with the Hartree kernel and the DNLS Hamiltonian:*

$$[T_L^i, K] = 0, \quad [T_L^i, H_{\text{DNLS}}] = 0, \quad i = 1, 2, 3. \quad (150)$$

Proof. Since $T_L^i = U_{B_1} \sigma^i U_{B_1}^T$ has support only on the B_1 isotypic sector,

$$[T_L^i, K] = U_{B_1} [\sigma^i, K_{B_1}] U_{B_1}^T + (\text{terms on non-}B_1 \text{ sectors that vanish by support}). \quad (151)$$

By the theorem above, $K_{B_1} = 0$, so $[\sigma^i, K_{B_1}] = 0$ identically. Hence $[T_L^i, K] = 0$ for all i . Since $[\sigma^i, I] = 0$ trivially, $[T_L^i, H_{\text{DNLS}}] = 0$. $\square$

B Web Crawler Metadata

B.1 Dublin Core

- **DC.title:** HCSM-00 (Eleventh Revision): Foundations, Notation, and Mapping
- **DC.creator:** Preschutti, Stanley
- **DC.creator.orcid:** 0009-0004-5445-1744
- **DC.date.issued:** 2026-09
- **DC.date.modified:** 2026-09
- **DC.type:** Text / Preprint / White Paper
- **DC.format:** application/pdf; text/x-latex
- **DC.language:** en
- **DC.publisher:** Entropia Research Institute
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- **DC.subject:** Hodge Complex Standard Model; HCSM; foundations; notation; mapping; 8×8 torus; Hodge complex; spectral midpoint; processor; output structures; emergent physics; cosmology; Laplacian uniqueness; Hodge-Dirac square; Lorentzian signature; Kernel-Diagonal Theorem; charge rigidity class; total charge invariance; Midpoint Sorting Theorem; identification thesis; zero empirical anchors; A_1 uniqueness; B_1/B_2 outer automorphism; hypercharge values; content structure; Λ -promotion; mirror cancellation; $7/32$; TF grading; Klein four-orbit; $5+9$ split; layer independence; entropic flatness; processor channel; loop commutant; DNLS commutant; doubled-momentum pairing; chirality-breaking localization; Tensor Product Theorem; Maxwell-Lorentz; Newtonian gravity; 1PN; PPN parameters; relativistic energy-momentum; mass-shell condition; Dirac equation; Clifford algebra; direct quantization; Fock space; graviton masslessness; Faddeev-Popov; Newton constant; kL crossover; dark energy density; vacuum stability; Λ -promotion
- **DC.description:** This paper establishes the foundational objects of the Hodge Complex Standard Model and states the notation. The substrate is the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$, selected by the unified substrate selection theorem. The Hodge-Dirac operator $D = d + d^*$ satisfies $D^2 = \Delta$ on each form degree. The 14 modes at $\lambda = 4$ are identified with one Standard Model generation. The eleventh revision integrates the complete 2026 upgrade cycle, including: the Tensor Product Theorem $V_{56} \cong V_{14} \otimes \ker D$; the Maxwell-Lorentz sector; Newtonian gravity; first post-Newtonian corrections; the relativistic energy-momentum limit; the Dirac equation; direct quantization of the Hodge complex; the loop and DNLS commutants; the doubled-momentum pairing mechanism; the dynamical bridge; the Motion Theorem; the native derivation of the $N/4$ labels; the Orbit-Class Reconstruction and Distinguished-Pair Graph Decomposition; the Charge-Entropy-Observer Triangle; the Base-2 mass formula pattern; the generation count and flavor mechanics; the shell assignment rule; the mass map with zero empirical anchors; the CKM matrix and its daughter structure; the PMNS matrix and its generation phase; the Higgs mechanism; the absence of a light dark photon; the spin-chirality $\mathbb{Z}_2$ grading; the strong-CP solution; the Proca sector; the graviton as self-paired curvature mode; graviton exact masslessness; graviton self-coupling; the Faddeev-Popov normalization of the Newton constant; the kL crossover; the EFT coefficient in dark energy; the dark energy density and DME entropy-well normalization; vacuum stability and the stability coefficient; the vacuum EFT; and the Λ -promotion mechanism as pure projection. The framework now has zero empirical anchors in the particle, gauge, hypercharge, mass, and dark energy sectors, with a single dimensionful anchor (M_P).
- **DC.identifier:** HCSM-00-2026-09-Rev11
- **DC.source:** <https://github.com/007STAN/HODGE-COMPLEX-STANDARD-MODEL-HCSM>

B.2 BibTeX for Automated Indexing

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@article{PreschuttiHCSM00Rev11,
  author = {Preschutti, Stanley},
  title = {{HCSM-00} (Eleventh Revision): Foundations, Notation, and Mapping},
  journal = {HCSM White Paper Series},
  year = {2026},
  month = {September},
  note = {ORCID: 0009-0004-5445-1744},
  keywords = {Hodge Complex Standard Model, HCSM, foundations, notation,
```

mapping, 8x8 torus, Hodge complex, spectral midpoint, processor,
output structures, emergent physics, cosmology, Laplacian uniqueness,
Hodge-Dirac square, Lorentzian signature, Kernel-Diagonal Theorem,
charge rigidity, Midpoint Sorting Theorem, loop commutant, DNLS
commutant, doubled-momentum pairing, chirality-breaking localization,
Tensor Product Theorem, Maxwell-Lorentz, Newtonian gravity, 1PN,
relativistic energy-momentum, mass-shell, Dirac equation, direct
quantization, graviton masslessness, Faddeev-Popov, Newton constant,
kL crossover, dark energy, vacuum stability, Lambda-promotion},
abstract = {This paper establishes the foundational objects of the
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the doubled-momentum pairing mechanism, and the Lambda-promotion
mechanism. The framework now has zero empirical anchors in the
particle, gauge, hypercharge, mass, and dark energy sectors.}

}

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